

Labs for ME 667 Mechatronics

Lab	Topic	Subtopic
1	Microcontroller Architecture	n/a
2	C Language, Microcontroller Programming, Communication	n/a
3	Digital I/O and ADC	n/a
4	Polling and Interrupt	n/a
5	Encoder	n/a
6	Logic Gates and Flip-Flops	n/a
7	555 Timer and Op Amp	555 timer, monostable and a-stable operation
7		Operational amplifier, inverting, noninverting, differential, summing
8	Interface with sensors	Infrared Range sensor
8		Temperature sensor
9	PWM interface with Actuators	Pulse width modulation with 2840 Board
9		low-pass filtered PWM
9		DC motor open-loop driving
9		Solenoid valve/relay
10	DC motor closed-loop Speed Control	DC motor modeling
9		Encoder for speed feedback
9		PWM for motor driving
9		Speed Controller Design
9		Implementation of Closed Speed Control

Homeworks

Homework	Chapter	Due
1	2	not recorded
2	4	not recorded
3	3	not recorded
4	6	not recorded
5	6	not recorded
6	6	not recorded
7	6	20-Oct
8	6,3	no
Exam1		24-Oct
9	6,9	3-Nov
10	8	7-Nov
11	Franklin ch4,5	
12	8,10	24-Nov
13	Franklin Ch8	1-Dec
14	Franklin Ch8	

Chapter 8 Important Topics

Digital control, Chapter 8

Topic	Page
DC gain	568
z pole table	566
difference equation	570

Laplace Transforms and z-Transforms of Simple Discrete-Time Functions

No.	$F(s)$	$f(kT)$	$F(z)$
1		$1, k = 0; 0, k \neq 0$	1
2		$1, k = k_0; 0, k \neq k_0$	z^{-k_0}
3	$\frac{1}{s}$	$1(kT)$	$\frac{z}{z-1}$
4	$\frac{1}{s^2}$	kT	$\frac{Tz}{(z-1)^2}$
5	$\frac{1}{s^3}$	$\frac{1}{2!}(kT)^2$	$\frac{T^2}{2} \left[\frac{z(z+1)}{(z-1)^3} \right]$
6	$\frac{1}{s^4}$	$\frac{1}{3!}(kT)^3$	$\frac{T^3}{6} \left[\frac{z(z^2+4z+1)}{(z-1)^4} \right]$
7	$\frac{1}{s^m}$	$\lim_{a \rightarrow 0} \frac{(-1)^{m-1}}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial a^{m-1}} e^{-akT} \right)$	$\lim_{a \rightarrow 0} \frac{(-1)^{m-1}}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial a^{m-1}} \frac{z}{z-e^{-aT}} \right)$
8	$\frac{1}{s+a}$	e^{-akT}	$\frac{z}{z-e^{-aT}}$
9	$\frac{1}{(s+a)^2}$	kTe^{-akT}	$\frac{Tze^{-aT}}{(z-e^{-aT})^2}$
10	$\frac{1}{(s+a)^3}$	$\frac{1}{2}(kT)^2 e^{-akT}$	$\frac{T^2}{2} e^{-aT} z \frac{(z+e^{-aT})}{(z-e^{-aT})^3}$
11	$\frac{1}{(s+a)^m}$	$\frac{(-1)^{m-1}}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial a^{m-1}} e^{-akT} \right)$	$\frac{(-1)^{m-1}}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial a^{m-1}} \frac{z}{z-e^{-aT}} \right)$
12	$\frac{a}{s(s+a)}$	$1 - e^{-akT}$	$\frac{z(1-e^{-aT})}{(z-1)(z-e^{-aT})}$
13	$\frac{a}{s^2(s+a)}$	$\frac{1}{a}(akT - 1 + e^{-akT})$	$\frac{z[(aT-1)+e^{-aT}]z + (1-e^{-aT}-aTe^{-aT})}{a(z-1)^2(z-e^{-aT})}$
14	$\frac{b-a}{(s+a)(s+b)}$	$e^{-akT} - e^{-bkT}$	$\frac{(e^{-aT} - e^{-bT})z}{(z-e^{-aT})(z-e^{-bT})}$
15	$\frac{s}{(s+a)^2}$	$(1 - akT)e^{-akT}$	$\frac{z\{z-e^{-aT}(1+aT)\}}{(z-e^{-aT})^2}$
16	$\frac{a^2}{s(s+a)^2}$	$1 - e^{-akT}(1 + akT)$	$\frac{z\{z(1-e^{-aT}-aTe^{-aT}) + e^{-2aT} - e^{-aT} + aTe^{-aT}\}}{(z-1)(z-e^{-aT})^2}$
17	$\frac{(b-a)s}{(s+a)(s+b)}$	$be^{-bkT} - ae^{-akT}$	$\frac{z\{z(b-a) - (be^{-aT} - ae^{-bT})\}}{(z-e^{-aT})(z-e^{-bT})}$
18	$\frac{a}{s^2+a^2}$	$\sin akT$	$\frac{z \sin aT}{z^2 - (2 \cos aT)z + 1}$
19	$\frac{s}{s^2+a^2}$	$\cos akT$	$\frac{z(z - \cos aT)}{z^2 - (2 \cos aT)z + 1}$
20	$\frac{s+a}{(s+a)^2+b^2}$	$e^{-akT} \cos bkT$	$\frac{z(z-e^{-aT} \cos bT)}{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}}$
21	$\frac{b}{(s+a)^2+b^2}$	$e^{-akT} \sin bkT$	$\frac{ze^{-aT} \sin bT}{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}}$
22	$\frac{a^2+b^2}{s[(s+a)^2+b^2]}$	$1 - e^{-akT}(\cos bkT + \frac{a}{b} \sin bkT)$	$\frac{z(Az+B)}{(z-1)\{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}\}}$ $A = 1 - e^{-aT} \cos bT - \frac{a}{b} e^{-aT} \sin bT$ $B = e^{-2aT} + \frac{a}{b} e^{-aT} \sin bT - e^{-aT} \cos bT$

$F(s)$ is the Laplace transform of $f(t)$, and $F(z)$ is the z-transform of $f(kT)$.

Note: $f(t) = 0$ for $t = 0$.

Laplace Transforms and z-Transforms of Simple Discrete-Time Functions

No.	$F(s)$	$f(kT)$	$F(z)$
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2		$1, k = k_0; 0, k \neq k_0$	z^{-k_0}
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8	$\frac{1}{s+a}$	e^{-akT}	$\frac{z}{z-e^{-aT}}$
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10	$\frac{1}{(s+a)^3}$	$\frac{1}{2}(kT)^2 e^{-akT}$	$\frac{T^2}{2} e^{-aT} z \frac{(z+e^{-aT})}{(z-e^{-aT})^3}$
11	$\frac{1}{(s+a)^m}$	$\frac{(-1)^{m-1}}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial a^{m-1}} e^{-akT} \right)$	$\frac{(-1)^{m-1}}{(m-1)!} \left(\frac{\partial^{m-1}}{\partial a^{m-1}} \frac{z}{z-e^{-aT}} \right)$
12	$\frac{a}{s(s+a)}$	$1 - e^{-akT}$	$\frac{z(1-e^{-aT})}{(z-1)(z-e^{-aT})}$
13	$\frac{a}{s^2(s+a)}$	$\frac{1}{a}(akT - 1 + e^{-akT})$	$\frac{z[(aT-1)+e^{-aT}]z + (1-e^{-aT}-aTe^{-aT})}{a(z-1)^2(z-e^{-aT})}$
14	$\frac{b-a}{(s+a)(s+b)}$	$e^{-akT} - e^{-bkT}$	$\frac{(e^{-aT} - e^{-bT})z}{(z-e^{-aT})(z-e^{-bT})}$
15	$\frac{s}{(s+a)^2}$	$(1 - akT)e^{-akT}$	$\frac{z\{z-e^{-aT}(1+aT)\}}{(z-e^{-aT})^2}$
16	$\frac{a^2}{s(s+a)^2}$	$1 - e^{-akT}(1 + akT)$	$\frac{z\{z(1-e^{-aT}-aTe^{-aT}) + e^{-2aT} - e^{-aT} + aTe^{-aT}\}}{(z-1)(z-e^{-aT})^2}$
17	$\frac{(b-a)s}{(s+a)(s+b)}$	$be^{-bkT} - ae^{-akT}$	$\frac{z\{z(b-a) - (be^{-aT} - ae^{-bT})\}}{(z-e^{-aT})(z-e^{-bT})}$
18	$\frac{a}{s^2+a^2}$	$\sin akT$	$\frac{z \sin aT}{z^2 - (2 \cos aT)z + 1}$
19	$\frac{s}{s^2+a^2}$	$\cos akT$	$\frac{z(z - \cos aT)}{z^2 - (2 \cos aT)z + 1}$
20	$\frac{s+a}{(s+a)^2+b^2}$	$e^{-akT} \cos bkT$	$\frac{z(z-e^{-aT} \cos bT)}{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}}$
21	$\frac{b}{(s+a)^2+b^2}$	$e^{-akT} \sin bkT$	$\frac{ze^{-aT} \sin bT}{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}}$
22	$\frac{a^2+b^2}{s[(s+a)^2+b^2]}$	$1 - e^{-akT}(\cos bkT + \frac{a}{b} \sin bkT)$	$\frac{z(Az+B)}{(z-1)\{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}\}}$ $A = 1 - e^{-aT} \cos bT - \frac{a}{b} e^{-aT} \sin bT$ $B = e^{-2aT} + \frac{a}{b} e^{-aT} \sin bT - e^{-aT} \cos bT$

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MEEN 433/667 Mechatronics

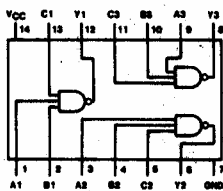
Handout #2

74 Series TTL & HCMOS Pin-Out

00 Quad 2 Input NAND 7400 74AS00 74HC00 74LS00 74ALS1000 74HCT00 74S00 74AS1000 40H000 74ALS00 74F00	01 Quad 2 Input NAND with OC 7401 74HC01 74LS01 74ALS01	02 Quad 2 Input NOR 7402 74AS02 74HC02 74LS02 74ALS1002 40H002 74ALS02 74AS1002 74F02 74S02	03 Quad 2 Input NAND with OC 7403 74ALS1003 74HC03 74LS03 74S03 74ALS03
04 Hex Inverter 7404 74AS04 74HCU04 74LS04 74ALS1004 74HCT04 74S04 74AS1004 74HCT04 74ALS04 40H004 74F04	05 Hex Inverter with OC 7405 74HCT05 74LS05 74ALS05 74S05 74ALS1005	06 Hex Inverter Buffer with OC High Voltage Output 7406 74LS06	07 Hex Buffer with OC High Voltage Output 7407 74HC07 74LS07
08 Quad 2 Input AND 7408 74AS08 74HC08 74LS08 74ALS1008 74HCT08 74S08 74AS1008 40H008 74ALS08 74F08	09 Quad 2 Input AND with OC 7409 74LS09 74S09 74ALS09	10 Triple 3 Input NAND 7410 74AS10 74HC10 74LS10 74ALS1010 74HCT10 74S10 40H010 74ALS10 74F10	11 Triple 3 Input AND 7411 74F11 74HC11 74LS11 74ALS1011 74HCT11 74S11 40H011 74ALS11 74AS11

12

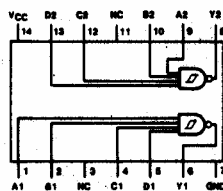
Triple 3 Input NAND with OC



7412
74LS12
74S12
74ALS12

13

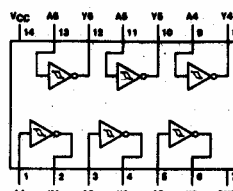
Dual 4 Input NAND Schmitt Trigger



7413 74HC13
74LS13
74F13

14

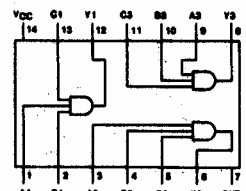
Hex inverter Schmitt Trigger



7414 74HC14
74LS14 74HCT14
74F14

15

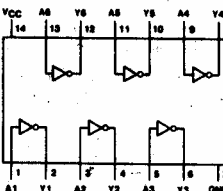
Triple 3 Input AND with OC



74LS15
74S15
74ALS15

16

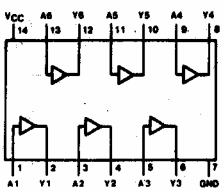
Hex Inverter Buffer 15V with OC



7416
74LS16

17

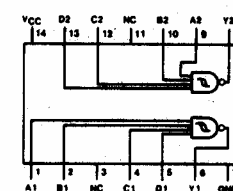
Hex Buffer 15V with OC



7417
74LS17

18

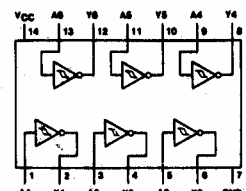
Dual 4 Input NAND Schmitt Trigger



74LS18

19

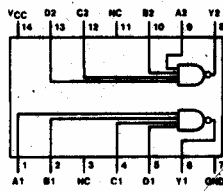
Hex Inverter Schmitt Trigger



74LS19

20

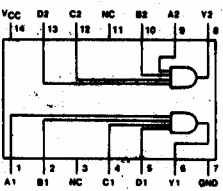
Dual 4 Input NAND



7420 74AS20 74HC20
74LS20 74ALS1020 74HCT20
74S20 40H020
74ALS20
74F20

21

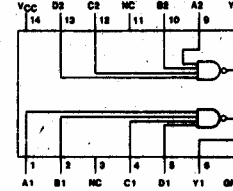
Dual 4 Input AND with OC



7421 74HC21
74LS21 40H021
74ALS21
74AS21

22

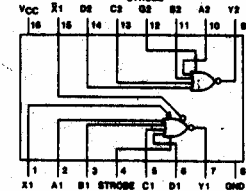
Dual 4 Input NAND with OC



7422 74ALS22
74LS22
74S22

23

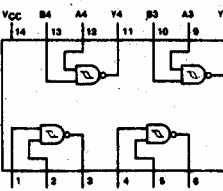
Expandable Dual 4 Input NOR with Strobe



7423

24

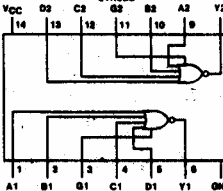
Quad 2 Input NAND Schmitt Trigger



74LS24

25

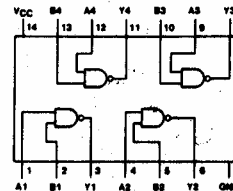
Dual 4 Input NOR with Strobe



7425

26

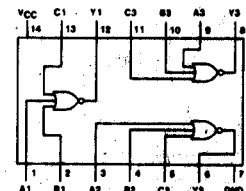
Quad 2 Input NAND 15V with OC



7426
74LS26

27

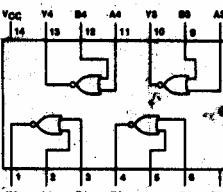
Triple 3 Input NOR



7427 74HC27
74LS27 74HCT27
74ALS27 40H027
74AS27

28

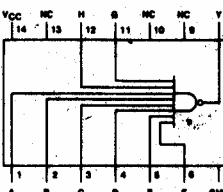
Quad 2 Input NOR Buffer



7428
74LS28
74ALS28
74ALS1028

30

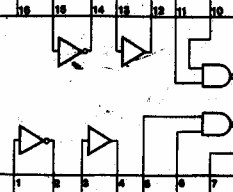
8 Input NAND



7430 74HC30
74LS30
74ALS30
74S30
74AS30

31

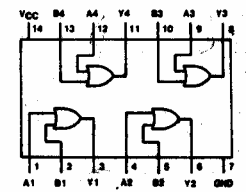
Delay Element



74LS31

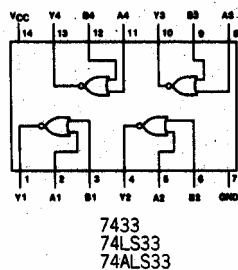
32

Quad 2 Input OR

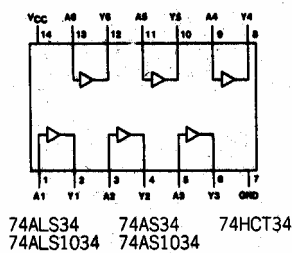


7432 74AS32 74HC32
74LS32 74ALS1032 74HCT32
74S32 74AS1032 40H032
74ALS32
74F32

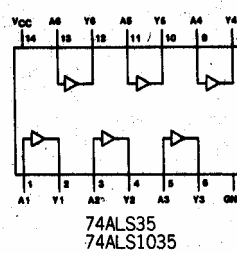
33
Quad 2 Input NOR Buffer with OC



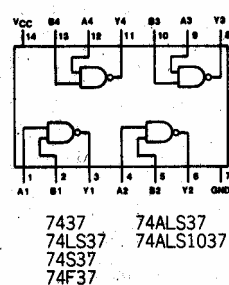
34
Hex Buffer



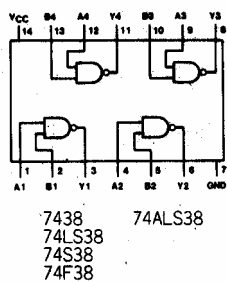
35
Hex Buffer with OC



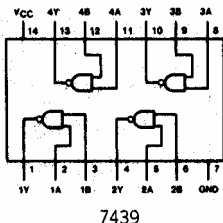
37
Quad 2 Input NAND Buffer



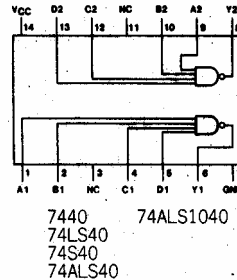
38
Quad 2 Input NAND Buffer with OC



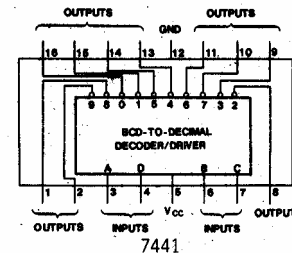
39
Quad 2 Input NAND Buffer with OC



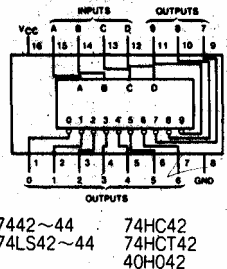
40
Dual 4 Input NAND Buffer



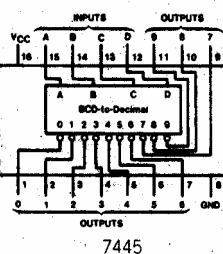
41
BCD to Decimal Decoder/Driver



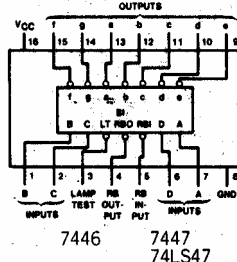
42:BCD to Decimal Decoder
43:Excess 3 to Decimal Decoder
44:Excess 3 Gray to Decimal Decoder



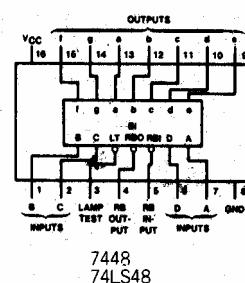
45
BCD to Decimal Decoder 30V with OC



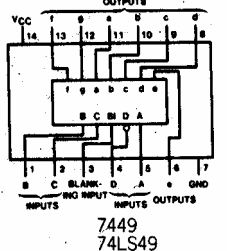
46:BCD to Seven Segment Decoder 30V with OC
47:BCD to Seven Segment Decoder 15V with OC



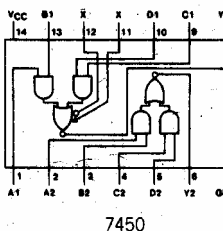
48
BCD to Seven Segment Decoder
2kΩ pullup with OC



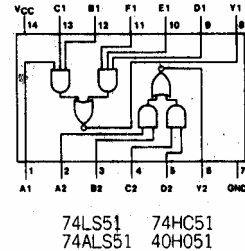
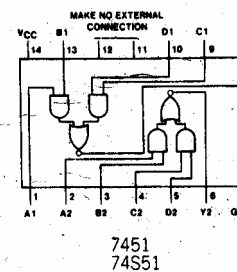
49
BCD to Seven Segment Decoder
5.5V with OC



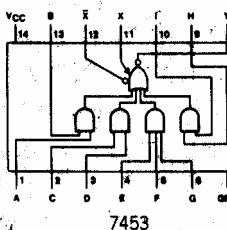
50
Dual 2 wide 2 Input AND OR INVERT



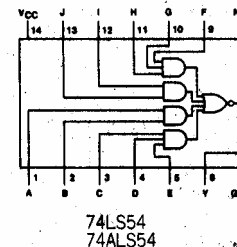
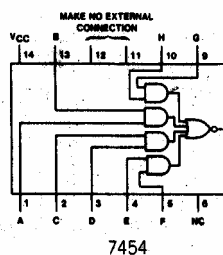
51
Dual 2 wide 2 Input AND OR INVERT



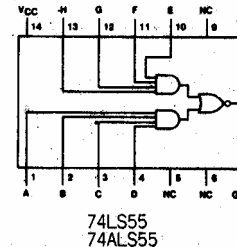
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4 wide 2 Input AND OR INVERT



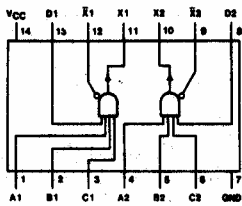
54
4 wide 2 Input AND OR INVERT



55
Expandable 2 wide 4 Input AND OR INVERT

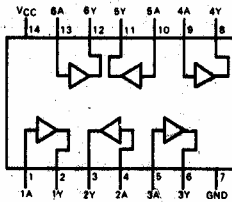


60 Dual 4 Input Expander



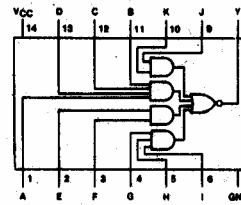
7460

63 Hex Current Sense Interface GATE



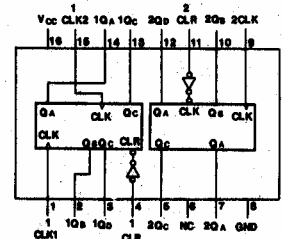
74LS63

64:4-2-3-2 Input AND OR INVERT 65:4-2-3-2 Input AND OR INVERT with OC



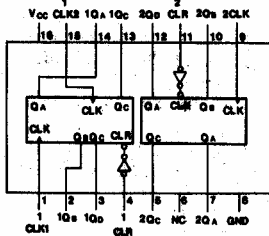
74S64 74S65
74F64 74ALS65
74ALS64

68 Dual Decade Counter



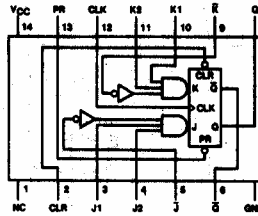
74LS68

69 Dual 4 Bit Binary Counter



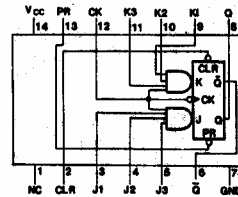
74LS69

70 AND-gated JK-FF with Preset and Clear



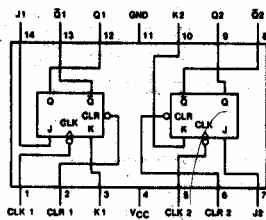
7470

72 AND-gated JK-FF with Preset and Clear



7472

73 Dual JK Flip Flop



7473 74HC73
74LS73 74HCT73
74ALS73

FUNCTION TABLE

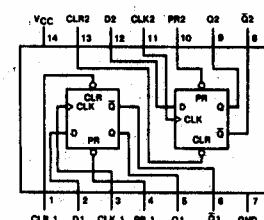
Inputs				Outputs	
CLR	CLK	J	K	Q	Q̄
L	X	X	X	L	H
H	X	L	L	Q0	Q0
H	X	H	L	H	L
H	X	L	H	L	H
H	X	H	H	TOGGLE	TOGGLE

7473

Inputs				Outputs	
CLR	CLK	J	K	Q	Q̄
L	X	X	X	L	H
H	X	L	L	Q0	Q0
H	X	H	L	H	L
H	X	L	H	L	H
H	X	H	H	TOGGLE	TOGGLE
H	X	X	X	Q0	Q0

74LS73

74 Dual D Type Flip Flop

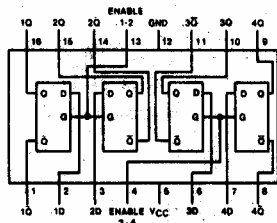


7474 74S74 74HC74
74LS74 74ALS74 74HCT74
74AS74 74F74 40H074

FUNCTION TABLE

Inputs				Outputs	
PR	CLR	CLK	D	Q	Q̄
L	H	X	X	L	L
H	L	X	X	L	H
L	L	X	X	H	H
H	H	X	X	Q0	Q0
H	H	L	L	H	L
H	H	L	H	L	H
H	H	X	X	TOGGLE	TOGGLE

75 4 Bit Latch

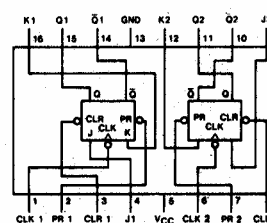


7475 74HC75
74LS75 74HCT75

FUNCTION TABLE (Each Latch)

Inputs		Outputs	
D	G	Q	Q̄
L	H	L	H
H	H	H	L
X	L	Q0	Q0

76 Dual JK Flip Flop



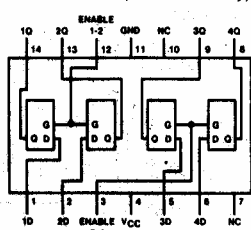
7476⁽¹⁾ 74HC76⁽¹⁾
74LS76⁽²⁾ 40H076⁽²⁾
74ALS76

FUNCTION TABLE

Inputs				Outputs	
PR	CLR	CLK	J K	Q	Q̄
L	H	X	X X	L	L
H	L	X	X X	L	H
L	L	X	X X	H	H
H	H	X	X X	Q0	Q0
H	H	L	L L	H	L
H	H	L	H H	L	H
H	H	X	X X	TOGGLE	TOGGLE

Inputs				Outputs	
PR	CLR	CLK	J K	Q	Q̄
L	H	X	X X	L	L
H	L	X	X X	L	H
L	L	X	X X	H	H
H	H	X	X X	Q0	Q0
H	H	L	L L	H	L
H	H	L	H H	L	H
H	H	X	X X	TOGGLE	TOGGLE

77 4 Bit Bistable Latch (Flat Pack Only)

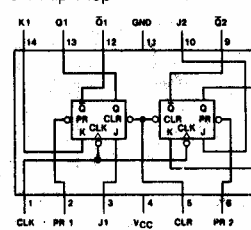


7477 74HC77
74ALS77

FUNCTION TABLE (Each Latch)

Inputs		Outputs	
D	G	Q	Q̄
L	H	L	H
H	H	H	L
X	L	Q0	Q0

78 Dual JK Flip Flop



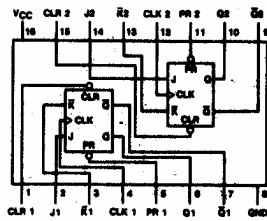
74LS78 74HC78
74ALS78

FUNCTION TABLE

Inputs				Outputs	
PR	CLR	CLK	J K	Q	Q̄
L	H	X	X X	L	L
H	L	X	X X	L	H
L	L	X	X X	H	H
H	H	X	X X	Q0	Q0
H	H	L	L L	H	L
H	H	L	H H	L	H
H	H	X	X X	TOGGLE	TOGGLE

80 Gated Full Adder 7480	81 16 Bit RAM (Open Collector) 7481	82 2 Bit Full Adder 7482	83 4 Bit Binary Full Adder 7483 74LS83
84 16 Bit RAM (Open Collector) 7484	85 4 Bit Comparator 7485 74LS85 74F85 74S85	86 Quad Exclusive OR 7486 74F86 74HC86 74LS86 74ALS86 74HCT86 74S86	89 64 Bit RAM (Open Collector) 7489 74LS89 74S89
90 Decade Counter 7490 74HC90 74LS90	91 8 Bit Shift Register SISO 7491 74HC91 74LS91	92 Divide by 12 Counter 7492 74HC92 74LS92	93 4 Bit Binary Counter 7493 74HC93 74LS93
94 4 Bit Shift Register 7494	95 4 Bit Shift Register PIPO 7495 74HC95 74LS95 74AS95	96 5 Bit Shift Register PIPO 7496 74LS96 74S96	97 6 Bit Synchronous Binary Rate Multiplier 7497
100 8 Bit Latch 74100	104 AND-gated JK-FF with Preset and Clear 74104	105 AND-gated JK-FF with Preset and Clear 74105	107 Dual JK Flip Flop (眞理値表は 73과 같다) 74107 74HC107 74LS107 74HCT107 74ALS107 40H107

109 Dual JK Flip Flop



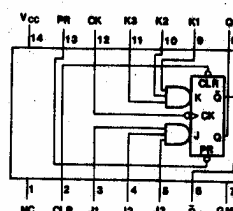
74109 74F109
74LS109 74S109
74ALS109 74AS109

FUNCTION TABLE

Inputs					Outputs	
PR	CLR	CLK	J	K	Q	Q̄
L	H	X	X	X	H	L
L	L	X	X	X	H*	H*
L	L	X	X	X	L	L
H	H	L	L	L	TOGGLE	TOGGLE
H	H	L	L	L	Q	Q̄
H	H	L	L	L	H	L
H	H	L	L	L	L	H

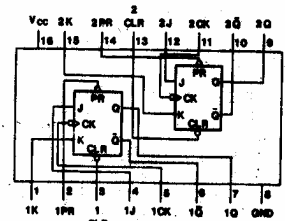
74HC109
74HCT109

110 AND-gated JK-FF with Preset and Clear



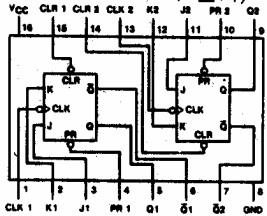
74110

111 Dual JK-FF with Preset and Clear



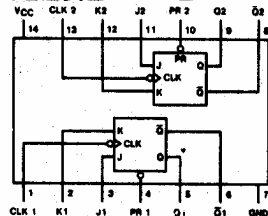
74111

112 Dual JK Flip Flop (眞理値表は LS76과 같다)



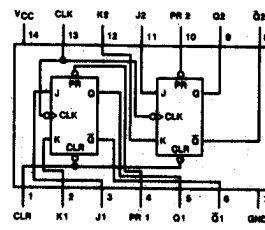
74LS112 74HC112
74ALS112 74HCT112
74S112
74F112
74AS112

113 Dual JK Flip Flop (眞理値表는 73과 같다)



74LS113 74HC113
74ALS113
74S113
74F113
74AS113

114 Dual JK Flip Flop

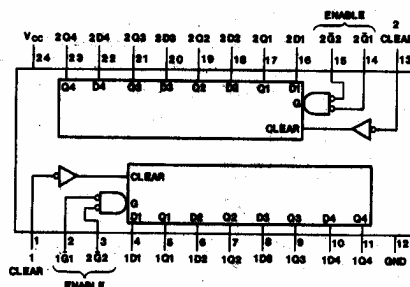


74LS114 74AS114 74HC114
74ALS114
74S114
74F114

FUNCTION TABLE

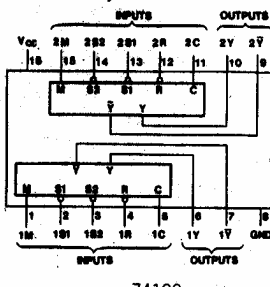
Inputs					Outputs	
PR	CLR	CLK	J	K	Q	Q̄
L	H	X	X	X	H	L
L	L	X	X	X	H*	H*
L	L	X	X	X	L	L
H	H	L	L	L	TOGGLE	TOGGLE
H	H	L	L	L	Q	Q̄
H	H	L	L	L	H	L
H	H	L	L	L	L	H

116 Dual 4 Bit Latch with Clear



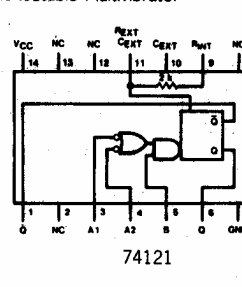
74116

120 Dual Pulse Synchronizer



74120

121 Monostable Multivibrator

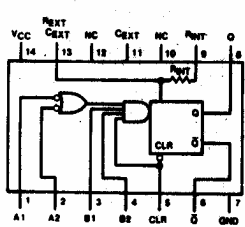


74121

FUNCTION TABLE

Inputs			Outputs	
A1	A2	B	Q	Q̄
L	X	H	L	H
X	L	H	L	H
X	X	L	L	H
X	X	H	L	H
L	L	H	L	H
L	L	L	L	H
L	L	L	L	H

122 Retriggerable Monostable Multivibrator

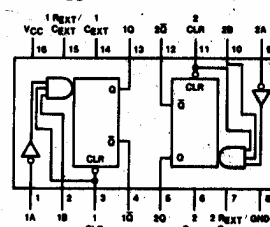


74122
74LS122

FUNCTION TABLE

Inputs					Outputs	
Clear	A1	A2	B1	B2	Q	Q̄
L	X	X	X	X	L	H
X	X	X	X	X	L	H
X	X	X	X	X	L	H
X	X	X	X	X	L	H
X	X	X	X	X	L	H
X	X	X	X	X	L	H
X	X	X	X	X	L	H
X	X	X	X	X	L	H

123 Dual Retriggerable Monostable Multivibrator

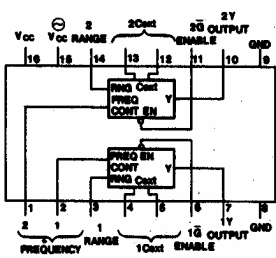


74123 74HC123
74LS123 74HCT123

FUNCTION TABLE

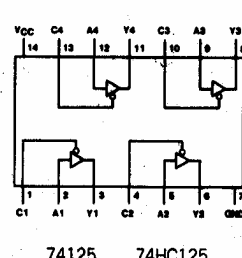
Inputs			Outputs	
Clear	A	B	Q	Q̄
L	X	X	L	H
X	X	X	L	H
X	X	X	L	H
X	X	X	L	H
X	X	X	L	H
X	X	X	L	H
X	X	X	L	H

124 Dual VCO



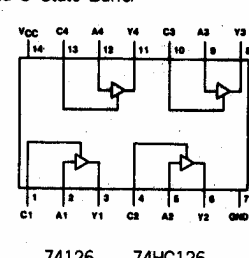
74LS124
74S124

125 Quad 3 State Buffer



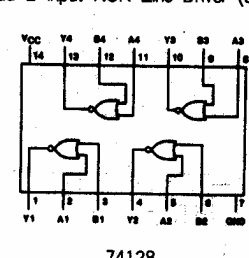
74125 74HC125
74LS125

126 Quad 3 State Buffer



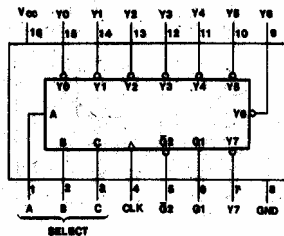
74126 74HC126
74LS126

128 Quad 2 Input NOR Line Driver (50 Ω)



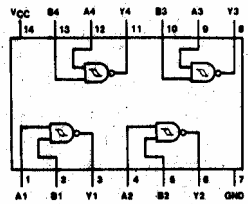
74128

131
3 to 8 Line Decoder with Address Latch



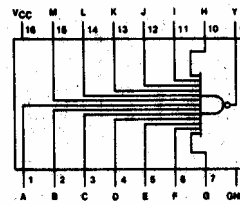
74ALS131 74HC131
74AS131

132
Quad 2 Input NAND Schmitt Trigger



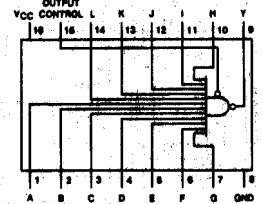
74132 74HC132
74ALS132 74HCT132
74S132
74F132

133
13 Input NAND



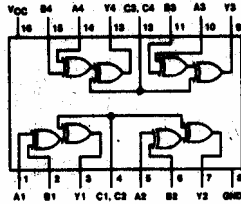
74LS133 74HC133
74ALS133
74S133

134
12 Input 3 State NAND



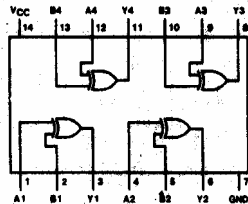
74S134

135
Quad Exclusive OR/NOR



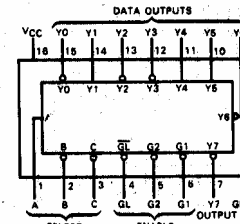
74S135

136
Quad Exclusive OR with OC



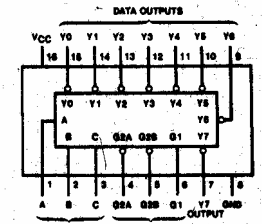
74136
74S136
74ALS136
74S136

137
3 to 8 Line Decoder/Latch



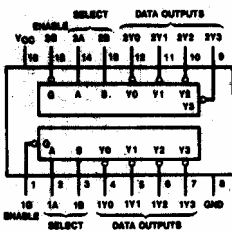
74LS137 74HC137
74S137 74HCT137
74ALS137
74S137
74AS137

138
3 to 8 Line Decoder



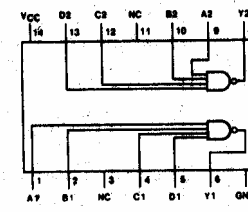
74LS138 74F138 74HC138
74S138 74AS138 74HCT138
74ALS138 40H138

139
Dual 2 to 4 Line Decoder



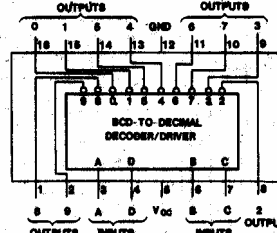
74LS139 74HC139
74S139 74HCT139
74ALS139 40H139
74AS139
74F139

140
Dual 4 Input NAND Line Driver (50Ω)



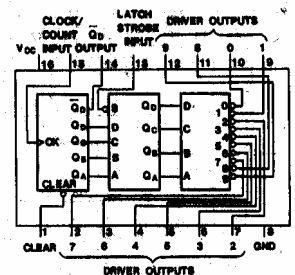
74S140

141
BCD to Decimal Decoder/Driver



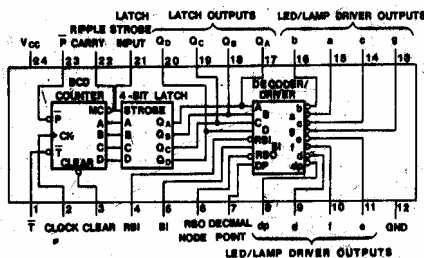
74141

142
BCD Counter/Latch/Decimal Decoder/Driver



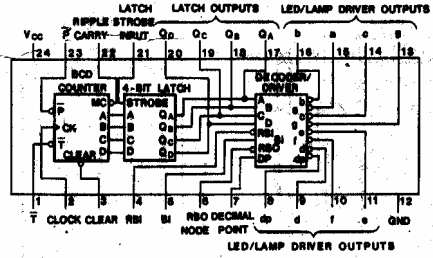
74142

143
BCD Counter/Latch/7 Segment Decoder/Driver



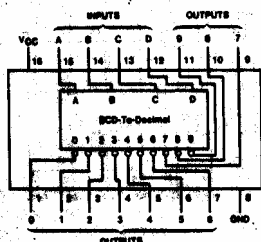
74143

144
BCD Counter/Latch/7 Segment Decoder/Driver



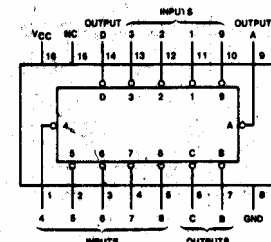
74144

145
BCD to Decimal Decoder 15V with OC



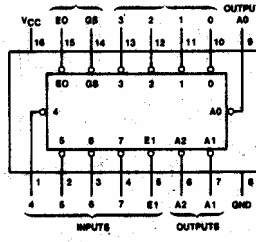
74145
74LS145

147
10 to 4 Line Priority Encoder



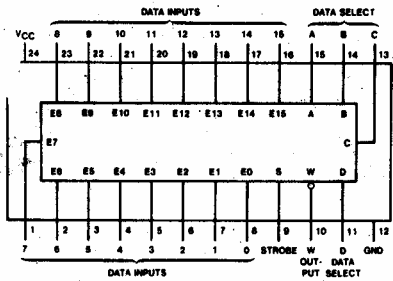
74147 74HC147
74S147 74HCT147
40H147

148
8 to 3 Line Priority Encoder



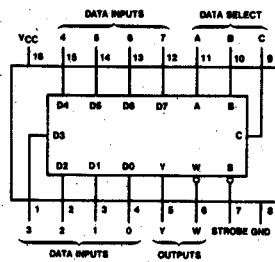
74148 74HC148
74S148 40H148
74F148
74HCT148

150
16 to 1 Line Data Selector



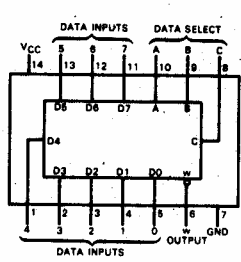
74150

151
8 to 1 Line Data Selector



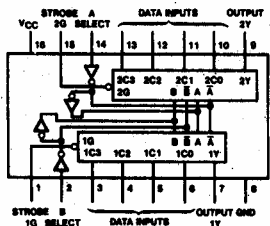
74151 74F151 74HC151
74LS151 74ALS151 74HCT151
74S151 74AS151 40H151

152
8 to 1 Line Data Selector



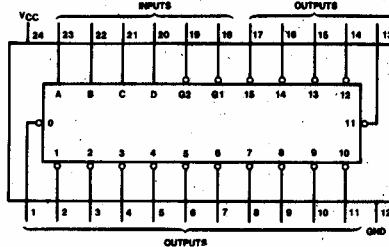
74152 74HC152
74LS152

153
Dual 4 to 1 Line Data Selector



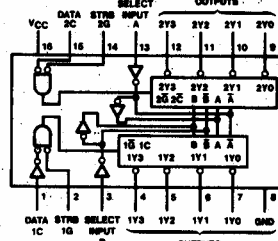
74153 74F153 74HC153
74LS153 74ALS153 74HCT153
74S153 74AS153 40H153

154
4 to 16 Line Decoder



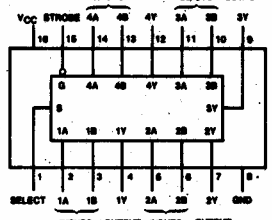
74154 74HC154
74LS154 74HCT154

155: Dual 2 to 4 Line Decoder
156: Dual 2 to 4 Line Decoder with OC



74155~156 40H155
74LS155~156 74HC155
74HCT155

157: Quad 2 to 1 Line Data Selector
158: Quad 2 to 1 Line Data Selector Inverting



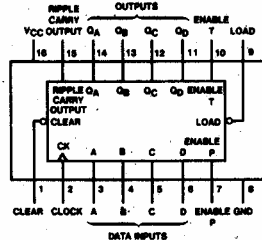
74157~158 74F157~158 74HC157~158
74LS157~158 74ALS157~158 40H157~158
74ALS157~158 74AS157~158 74HCT157~158

159
4 to 16 Line Decoder with OC

74154의 오픈 콜렉터형
(핀배치는 74154와 같다)

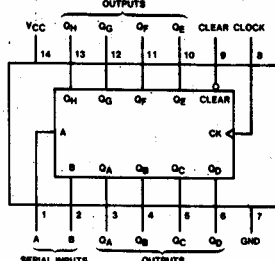
74159

160, 162: Synchronous Decoder Counter
161, 163: Synchronous Binary Counter



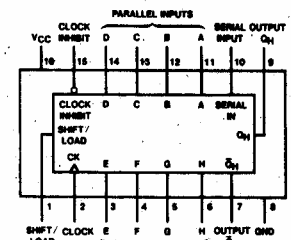
74160~163 74S162~163
74LS160~163 74HC160~163
74ALS160~163 40H160~163
74F160~163 74HCT160~163

164
8 Bit Shift Register



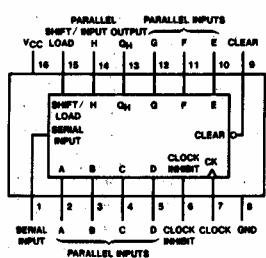
74164 74F164 74HC164
74LS164 74ALS164 74HCT164
40H164

165
8 Bit Shift Register PISO



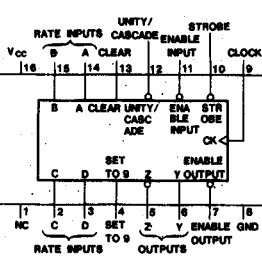
74165 74HC165
74LS165 74HCT165
74ALS165

166
8 Bit Shift Register PISO



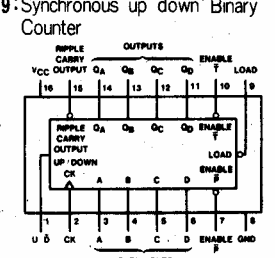
74166 40H166
74LS166 74HC166
74ALS166 74HCT166

167
BCD Synchronous Rate Multiplier



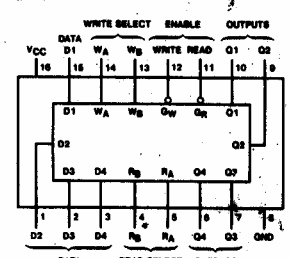
74167

168: Synchronous up down Decade Counter
169: Synchronous up down Binary Counter



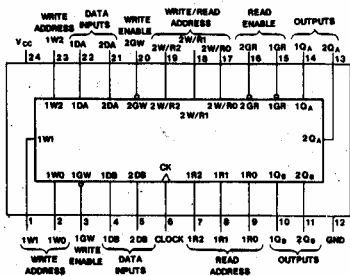
74LS168~169 74S168~169
74F168~169 74ALS168~169
74AS168~169

170
4 Word x 4 Bit Register File with OC



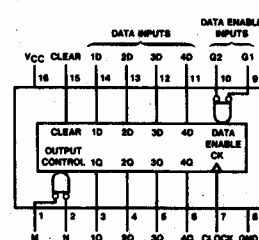
74170
74LS170

172
16 Bit Multiple-port Register File (3 state)



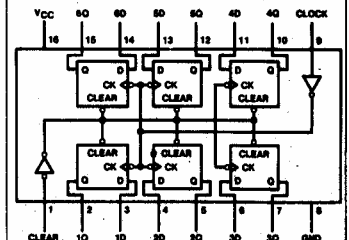
74172
74S172

173
Quad D Type Register (3 state)

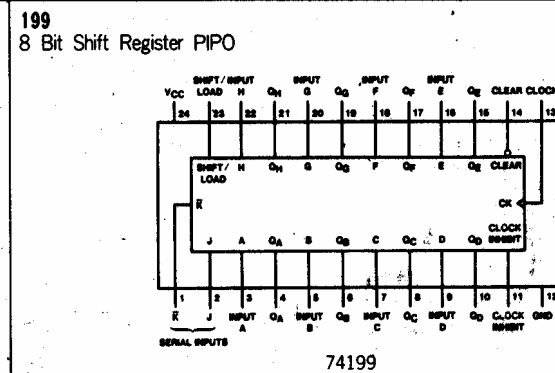
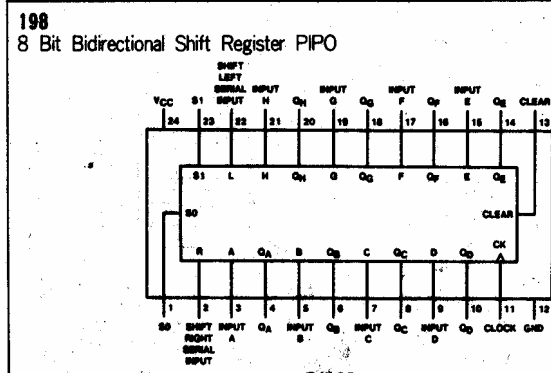
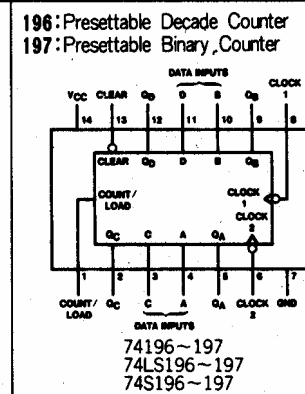
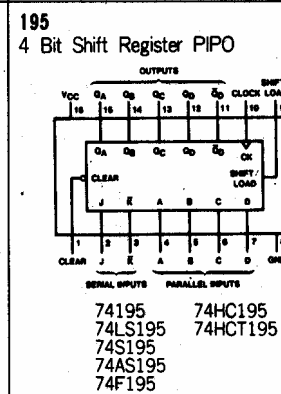
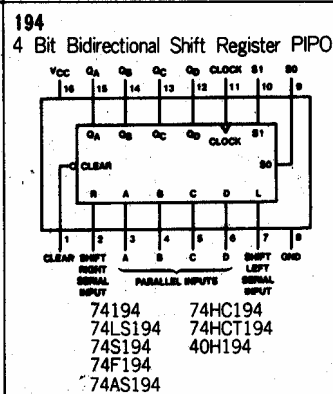
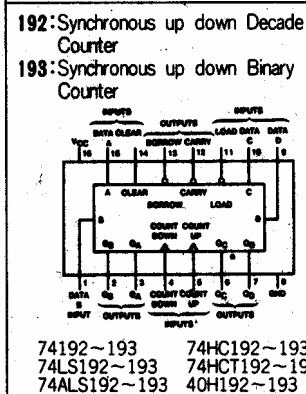
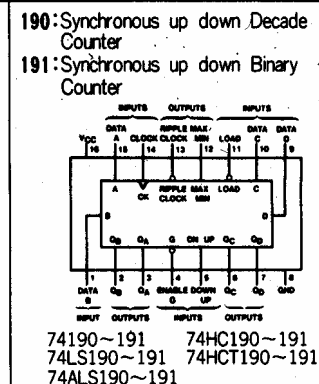
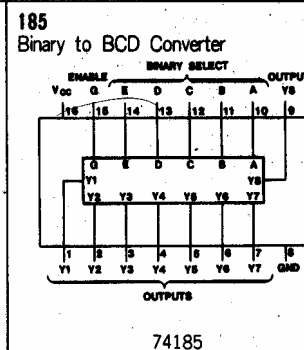
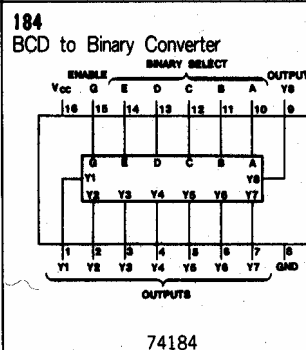
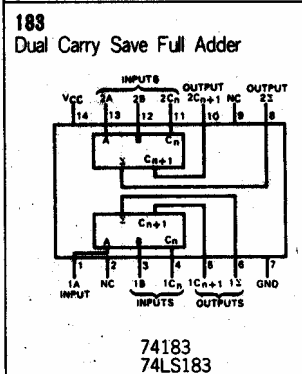
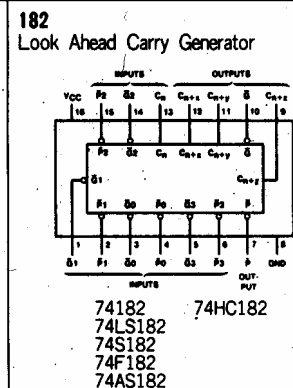
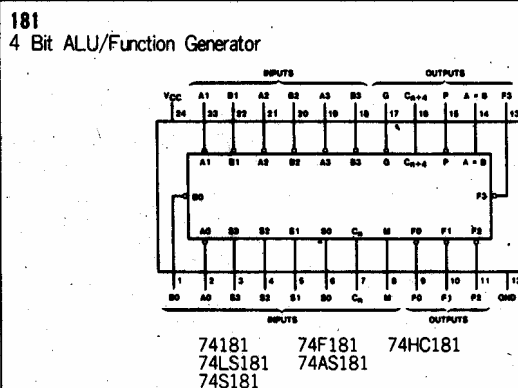
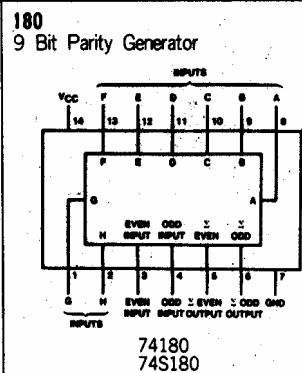
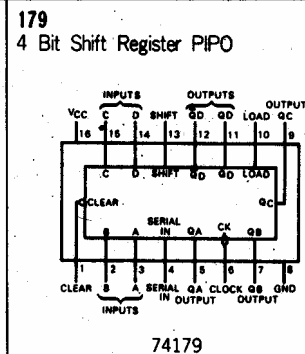
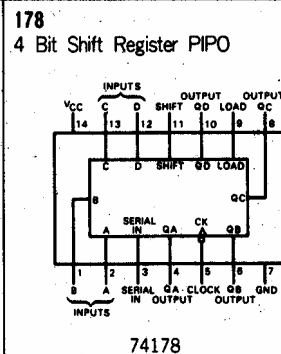
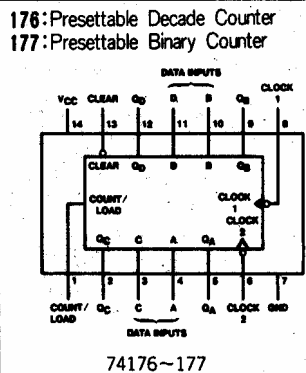
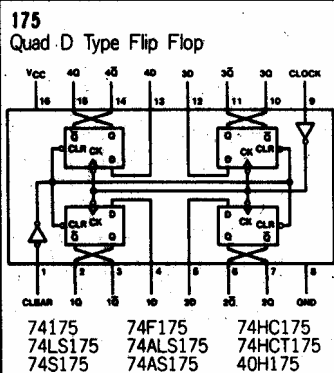


74173 74HC173
74LS173 74HCT173

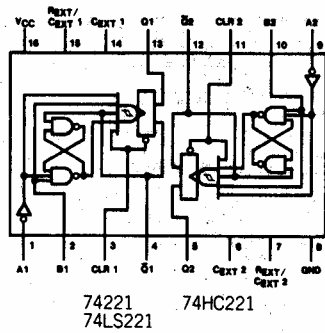
174
Hex D Type Flip Flop



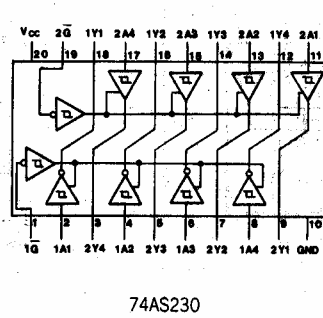
74174 74ALS174 74HC174
74LS174 74AS174 74HCT174
74F174 40H174



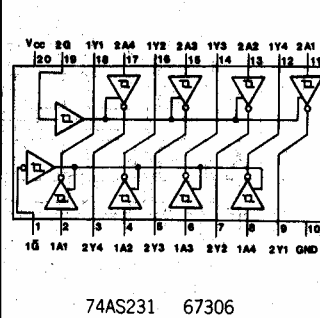
221 Dual Monostable Multivibrator



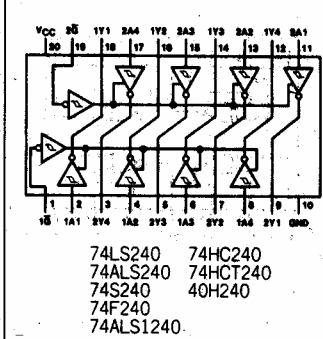
230 Octal 3 state Bus Driver



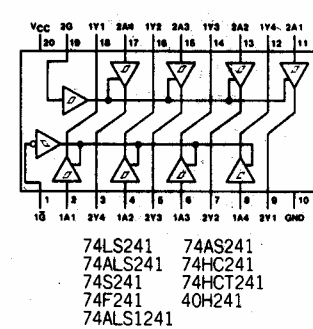
231 Octal 3 state Bus Inverter



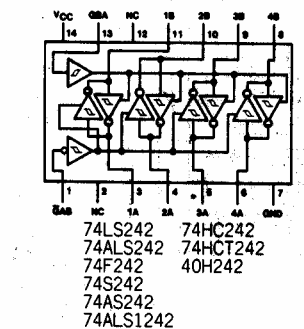
240 Octal 3 state Bus Inverter



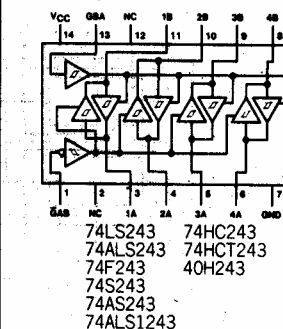
241 Octal 3 state Bus Driver



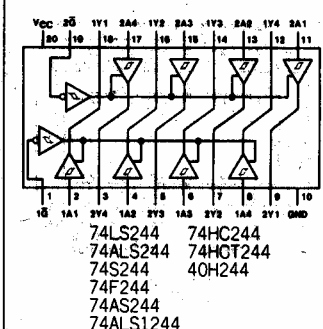
242 Quad 3 state Bus Transceiver



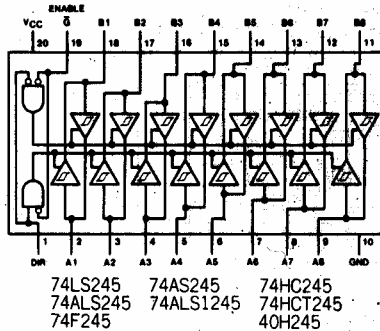
243 Quad 3 state Bus Transceiver



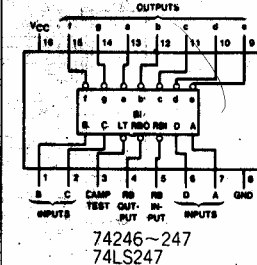
244 Octal 3 state Bus Driver



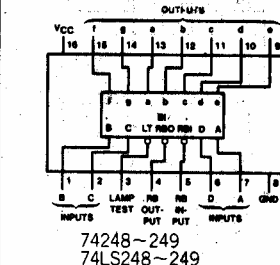
245 Octal 3 state Bus Transceiver



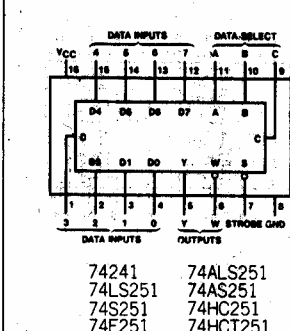
246:BCD to Seven Segment Decoder 30V with OC 247:BCD to Seven Segment Decoder 15V with OC



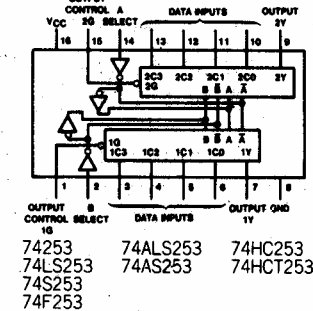
248:BCD to Seven Segment Decoder 2kΩ pullup with OC 249:BCD to Seven Segment Decoder 5.5V with OC



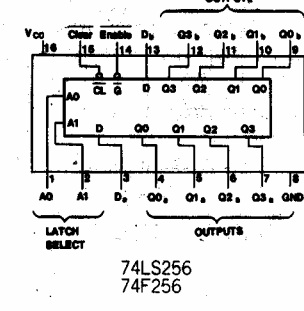
251 8 to 1 Line Data Selector 3 State



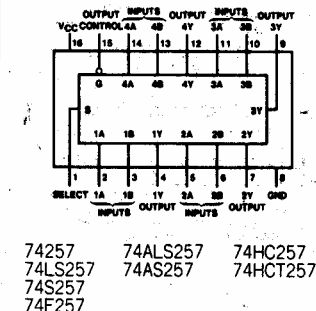
253 Dual 4 to 1 Line Data Selector 3 State



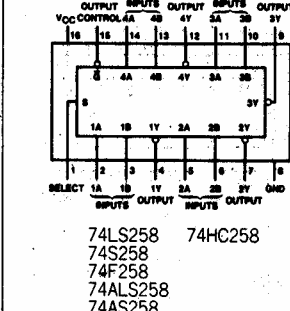
256 Dual 4 Bit Addressable Latch



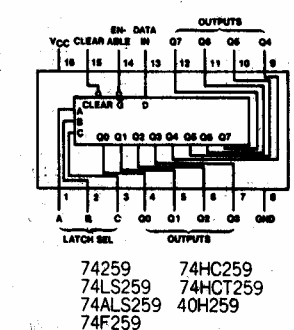
257 Quad 2 to 1 Data Selector 3 State



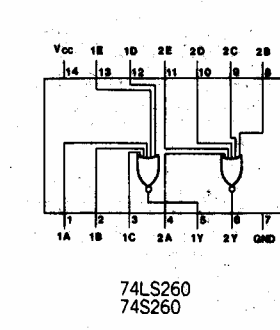
258 Quad 2 to 1 Data Selector 3 State



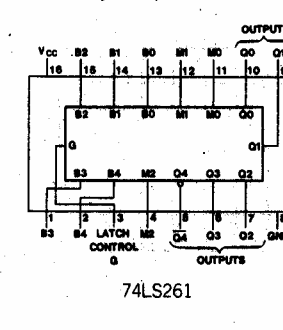
259 8 Bit Addressable Latch



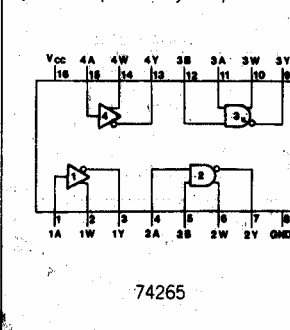
260 Dual 5 Input NOR



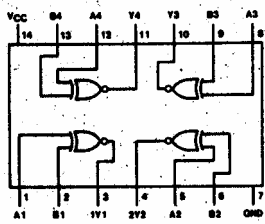
261 2x4 Binary Multiplier



265 Quad Complementary Output

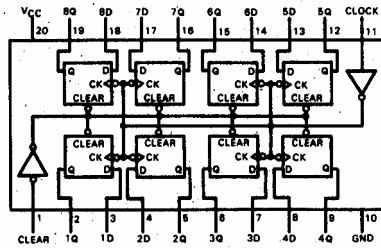


266
Quad 2 Input Exclusive NOR with OC



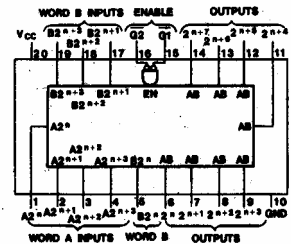
74LS266 74HC266

273
Octal D Type Flip Flop with Clear



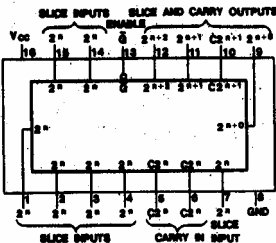
74273 74ALS273 74HC273
74LS273 74F273 40H273

274
4x4 Binary Multiplier



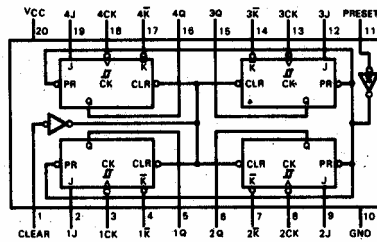
74S274

275
7 Bit Wallace Tree



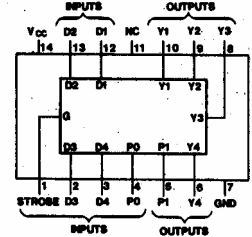
74LS275
74S275

276
Quad JK Flip Flop



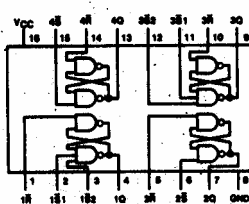
74276

278
4 Bit Cascadable Priority Register



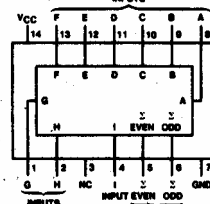
74278

279
Quad RS Latch



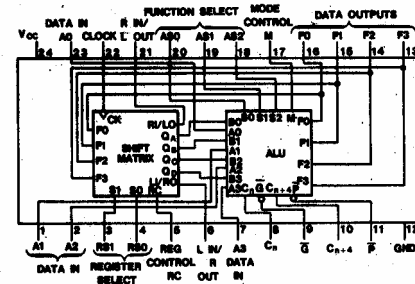
74279 74HC279
74LS279

280
9 Bit Parity Generator/Checker



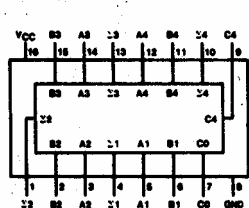
74LS280 74HC280
74S280 74HCT280
74F280
74ALS280
74AS280

281
4 Bit Binary Accumulator



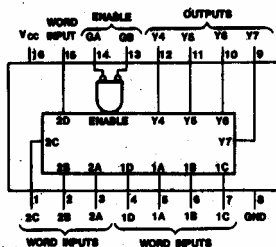
74LS281
74S281

283
4 Bit Binary Full Adder



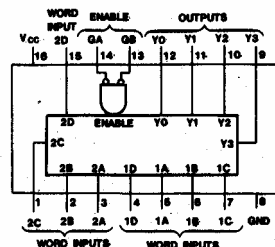
74283 74F283 74HC283
74LS283

284
4x4 Binary Multiplier



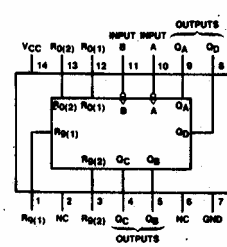
74284

285
4x4 Binary Multiplier



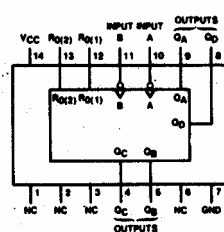
74285

290
Decade Counter



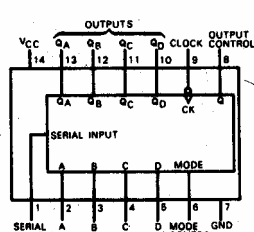
74290
74LS290

293
4 Bit Binary Counter



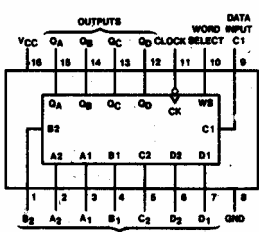
74293
74LS293

295
4 Bit Shift Register PIPO 3 State



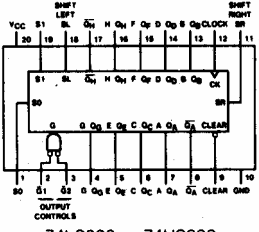
74LS295

298
Quad 2 Input Multiplexer with Storage



74298 74F298 74HC298
74LS298 74AS298

299
8 Bit Universal Shift/Storage Register



74LS299 74HC299
74S299 74HCT299
74F299
74S299

74LS322
74F322

Pin diagram of the 74ALS32 8-input OR gate. The diagram shows a rectangular package with pins numbered 1 to 14. Pin 1 is VCC, pin 14 is GND. Pins 2, 3, 4, 5, 6, 7, 8, 9 are inputs labeled G1 through G8. Pin 10 is the output labeled F. Pin 11 is labeled 'OR'. Pin 12 is labeled 'CLEAR'. Pin 13 is labeled 'SHIFT RIGHT'. Pin 14 is labeled 'SHIFT LEFT'. The diagram also shows the internal logic structure of the OR gate, including a network of inverters and OR gates.

Pin diagram of the 74LS347 decoder. The chip has 16 pins. Pin 16 is VCC. Pins 15, 14, 13, 12, 11, 10, 9 are outputs labeled F, E, D, C, B, A, and G respectively. Pins 1, 2, 3, 4, 5, 6 are inputs labeled B, C, LAMP TEST OUT, RB, and RA respectively. Pins 8 and 7 are inputs labeled D and A respectively. Pin 0 is GND. The chip is labeled 74LS347.

Pin diagram of the 74LS348 8421 BCD decoder. The chip has 16 pins. VCC is at pin 16 and GND is at pin 8. Inputs are pins 1-7 (labeled 1-7, 8-14, 15-21) and outputs are pins 2-7 (labeled 2-7, 8-14, 15-21). The diagram shows the internal logic structure with various gates and flip-flops.

V_{CC} 15 V_0 12 V_1 14 $\overline{OE}$ 13 V_2 12 V_3 11 V_0 10 V_1 9
 15 12 14 13 12 11 10 9
 15 12 14 13 12 11 10 9 GND
 74S350
 74F350

74LS353
74F353
74ALS353
74AS353

74365 74HC365
74LS365 74HCT365
74F365 40H365
74S365
74ALS365

The diagram shows a 4-bit ripple-carry adder circuit. It consists of four 4-bit comparators (74366, 74LS366, 74ALS366, 74F366, 74S366) connected in series. The inputs are labeled A1, A2, A3, A4 and B1, B2, B3, B4. The outputs are labeled Y1, Y2, Y3, Y4. The circuit is powered by VCC and GND. The output of the first comparator (Y1) is connected to the input of the second (A1). The output of the second (Y2) is connected to the input of the third (A2). The output of the third (Y3) is connected to the input of the fourth (A3). The output of the fourth (Y4) is connected to the input of the first (A1). The output of the first (Y1) is also connected to the input of the second (B1). The output of the second (Y2) is connected to the input of the third (B2). The output of the third (Y3) is connected to the input of the fourth (B3). The output of the fourth (Y4) is connected to the input of the first (B1). The output of the first (Y1) is also connected to the input of the second (A1). The output of the second (Y2) is connected to the input of the third (A2). The output of the third (Y3) is connected to the input of the fourth (A3). The output of the fourth (Y4) is connected to the input of the first (A1).

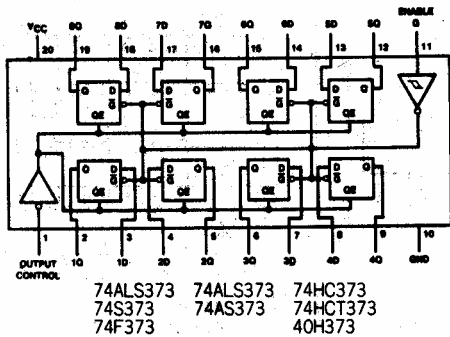
The diagram shows the internal logic of a 4-bit shift register. It consists of four D-type flip-flops (labeled 1, 2, 3, 4) and four 2-input AND gates. The clock input (G1) is connected to the clock inputs of all flip-flops. The data inputs are connected as follows: Input 1 (G1) is connected to the D input of flip-flop 1. Input 2 (A1) is connected to the D input of flip-flop 2. Input 3 (Y1) is connected to the D input of flip-flop 3. Input 4 (A7) is connected to the D input of flip-flop 4. The outputs of the flip-flops are connected to the inputs of the AND gates. The outputs of the AND gates are connected to the D inputs of the flip-flops. The output of the first AND gate is connected to the D input of flip-flop 1. The output of the second AND gate is connected to the D input of flip-flop 2. The output of the third AND gate is connected to the D input of flip-flop 3. The output of the fourth AND gate is connected to the D input of flip-flop 4. The output of the first AND gate is also connected to the output of flip-flop 1. The output of the second AND gate is also connected to the output of flip-flop 2. The output of the third AND gate is also connected to the output of flip-flop 3. The output of the fourth AND gate is also connected to the output of flip-flop 4. The output of the first AND gate is also connected to the output of flip-flop 2. The output of the second AND gate is also connected to the output of flip-flop 3. The output of the third AND gate is also connected to the output of flip-flop 4. The output of the fourth AND gate is also connected to the output of flip-flop 1.

74367	74HC367
74LS367	74HCT367
74ALS367	40H367
74F367	
74S367	

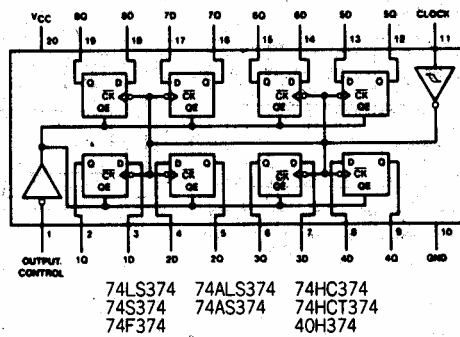
Pin diagram of the 74ALS368 8-bit bus driver. The diagram shows a 16-pin package with pins 1 through 16. Pin 1 is GND, pin 16 is Vcc. Pins 2, 4, 6, 8, 10, 12, 14, and 15 are inputs. Pins 3, 5, 7, 9, 11, 13, and 16 are outputs. The internal circuitry shows a series of inverters and NAND gates connecting the inputs to the outputs.

74368	74HC368
74ALS368	74HCT368
74ALS368	40H368
74F368	
74S368	

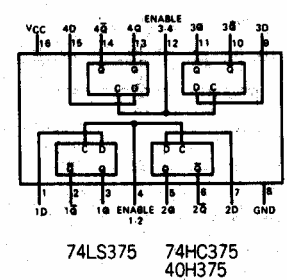
373
Octal Transparent Latch 3 state



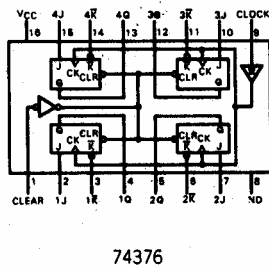
374
Octal D Type Flip Flop 3 state



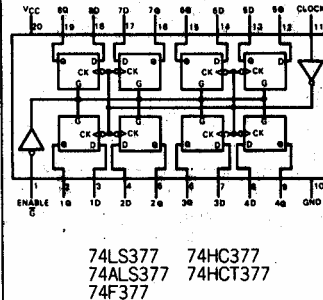
375
4 Bit Latch



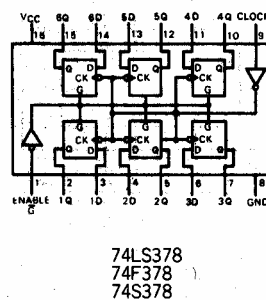
376
Quad JK Flip Flop



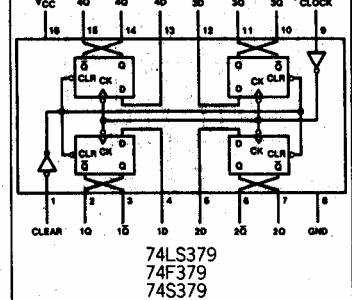
377
Octal D Type Flip Flop with Enable



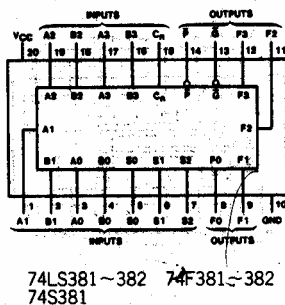
378
Hex D Type Flip Flop with Enable



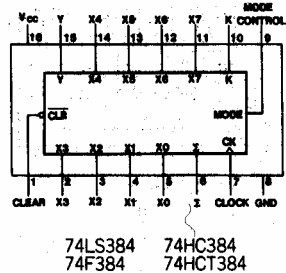
379
4 Bit D Type Flip Flop with Enable



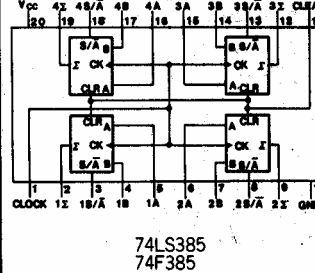
381, 382
ALU/Function Generator



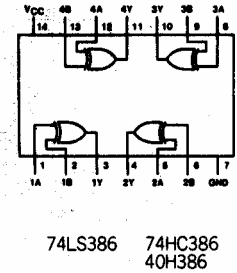
384
8 Bit by 1 Bit 2's Complement Multiplier



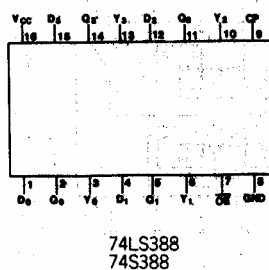
385
Quad Serial Adder/Subtractor



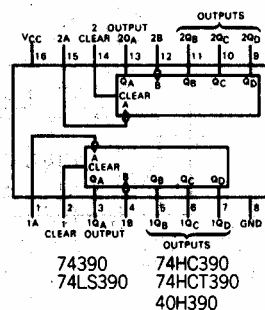
386
Quad Exclusive OR



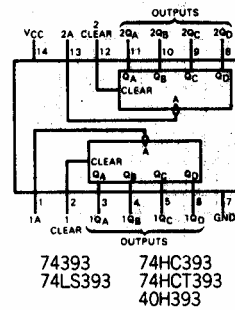
388
Quad D Type Flip Flop 3 state



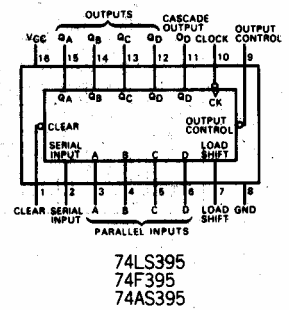
390
Dual Decade Counter



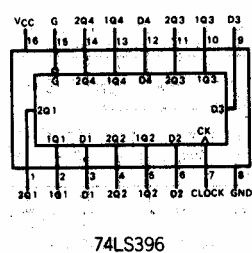
393
Dual 4 Bit Binary Counter



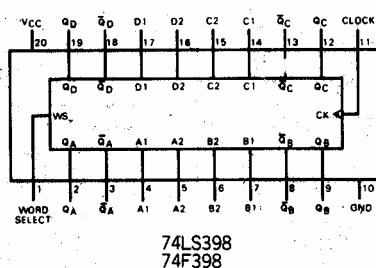
395
4 Bit Shift Register PIPO



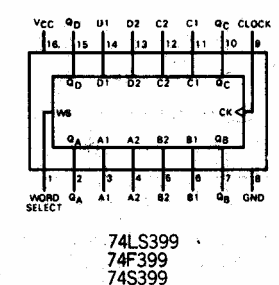
396
Octal Storage Register



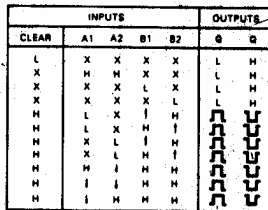
398
4 Bit 2 Input Multiplexer with Storage



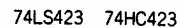
399
4 Bit 2 Input Multiplexer with Storage



422



428 Dual Monostable Multivibrator



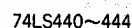
A24



426



440~444



445



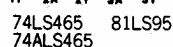
447



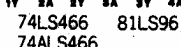
448



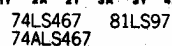
465



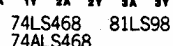
46



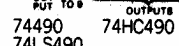
467



468



490



502



503

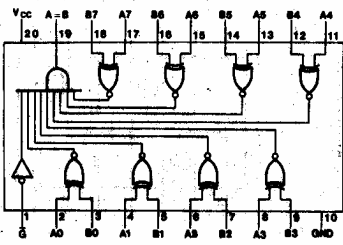


504



518

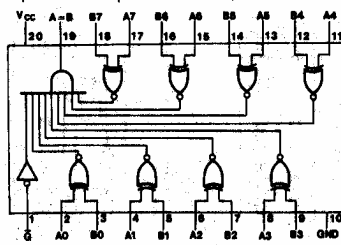
8 Bit Equal to Comparator with OC



74ALS518

519

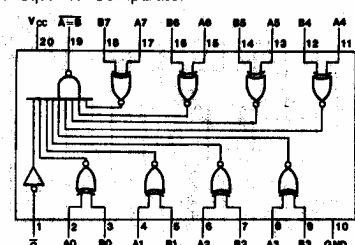
8 Bit Equal to Comparator with OC



74ALS519

520

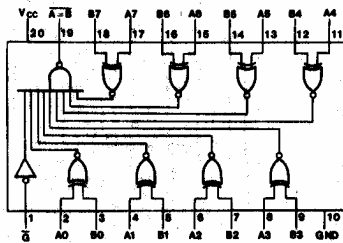
8 Bit Equal to Comparator



74ALS520

521

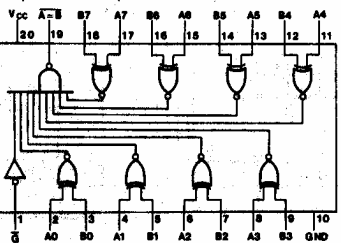
8 Bit Equal to Comparator



74ALS521 25LS2521
74ALS521
74F521

522

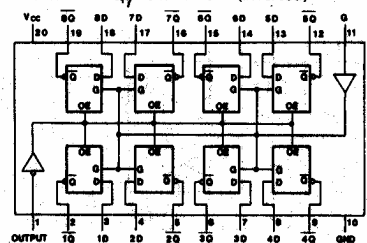
8 Bit Equal to Comparator with OC



74ALS522

533

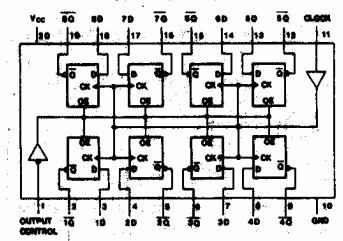
Octal 3 state Transparent Latch (Inverted)



74ALS533 74HC533
74ALS533 74HCT533
74S533
74F533

534

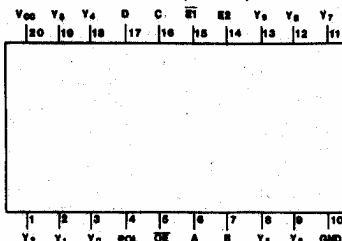
Octal 3 state D Type Flip Flop (Inverted)



74ALS534 74HC534
74ALS534 74HCT534
74S534
74AS534
74F534

537

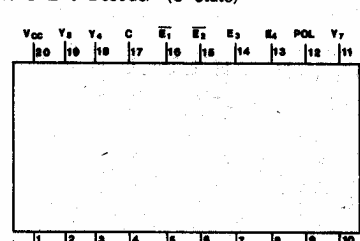
BCD to Decimal Decoder (3 state)



74ALS537 25LS2537
74ALS537
74F537

538

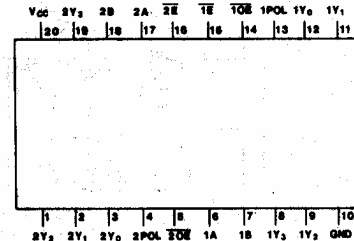
3 to 8 Line Decoder (3 state)



74ALS538 25LS2538
74ALS538
74F538

539

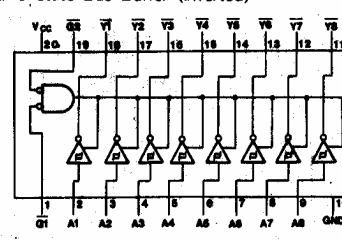
Dual 2 to 4 Line Decoder (3 state)



74ALS539 25LS2539
74ALS539
74F539

540

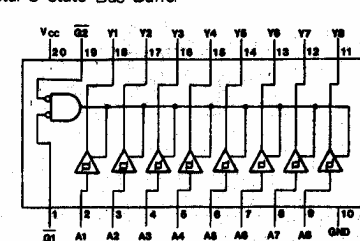
Octal 3 state Bus Buffer (Inverted)



74ALS540 74HC540
74ALS540 74HCT540

541

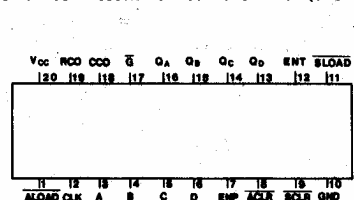
Octal 3 state Bus Buffer



74ALS541 74HC541
74ALS541 74HCT541

560

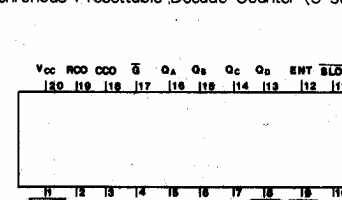
Synchronous Presettable Decade Counter (3 state)



74ALS560

561

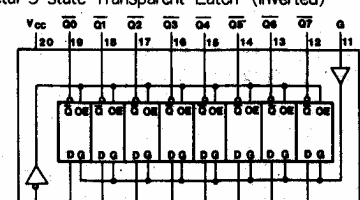
Synchronous Presettable Decade Counter (3 state)



74ALS561

563

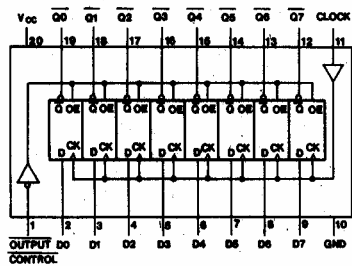
Octal 3 state Transparent Latch (Inverted)



74ALS563 74HC563
74ALS563 74HCT563
74S563

564

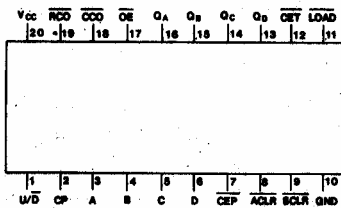
Octal 3 state D Type Flip Flop (Inverted)



74LS564 74HC564
74ALS564 74HCT564

568

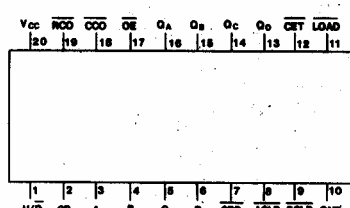
Synchronous Presettable Up/Down Counter (3 state)



74LS568 74F568
74ALS568

569

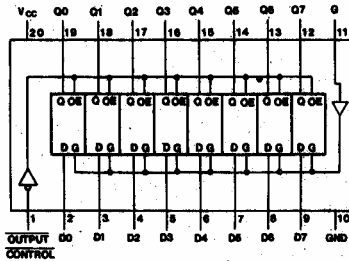
Synchronous Presettable Up/Down Binary Counter (3 state)



74LS569
74ALS569
74F569

573

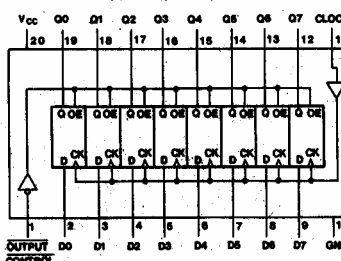
Octal 3 state Transparent Latch



74LS573 74HC573
74ALS573 74HCT573
74S573
74AS573

574

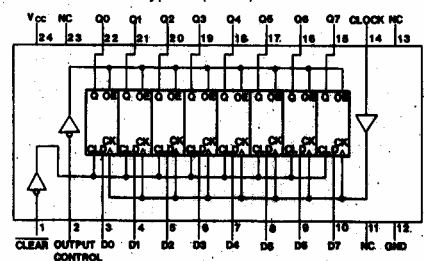
Octal 3 state D Type Flip Flop



74LS574 74HC574
74ALS574 74HCT574
74AS574

575

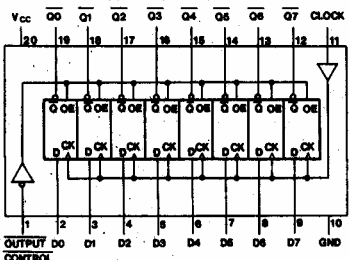
Octal 3 state D Type Flip Flop with Clear



74ALS575
74AS575

576

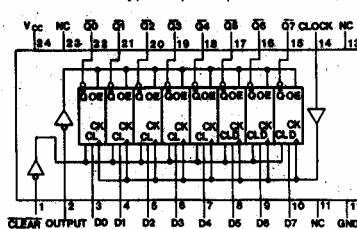
Octal 3 state D Type Flip Flop (Inverted)



74LS576 74HC576
74ALS576 74HCT576
74AS576

577

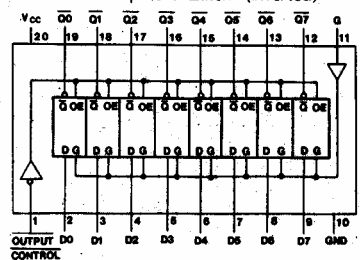
Octal 3 state D Type Flip Flop with Clear (Inverted)



74ALS577
74AS577

580

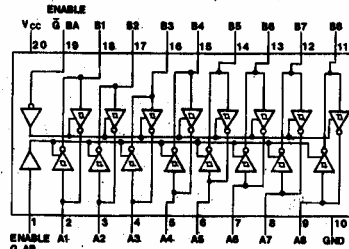
Octal 3 state Transparent Latch (Inverted)



74LS580 74HC580
74ALS580 74HCT580
74S580
74AS580

620

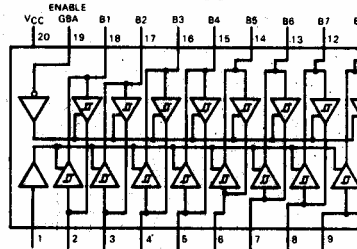
Octal 3 state Bus Transceiver (Inverted)



74LS620
74ALS620
74F620
74AS620

621

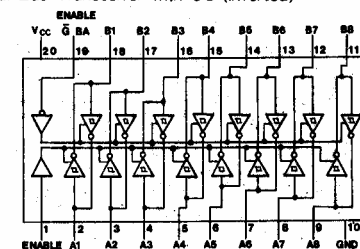
Octal Bus Transceiver with OC



74LS621
74ALS621
74F621
74AS621

622

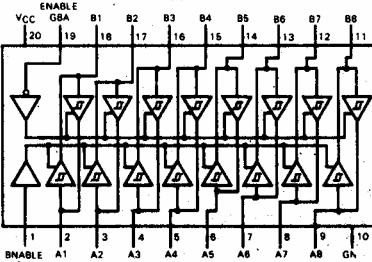
Octal Bus Transceiver with OC (Inverted)



74LS622
74ALS622
74F622
74AS622

623

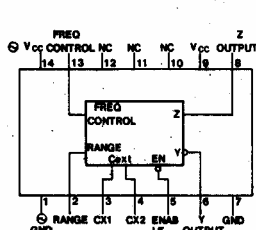
Octal 3 state Bus Transceiver



74LS623
74ALS623
74F623
74AS623

624

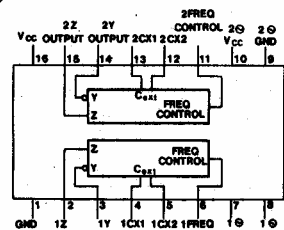
VCO



74LS624

625

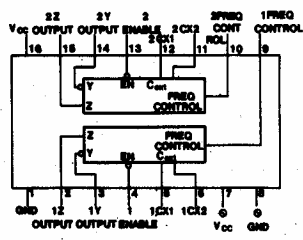
Dual VCO



74LS625

626

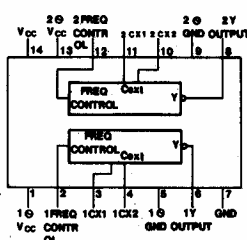
Dual VCO



74LS626

627

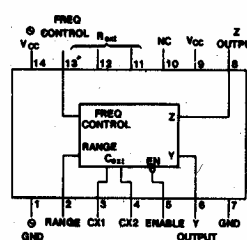
Dual VCO



74LS627

628

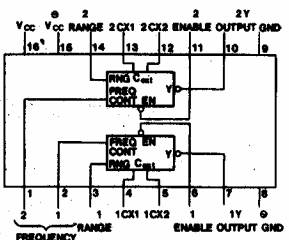
VCO



74LS628

629

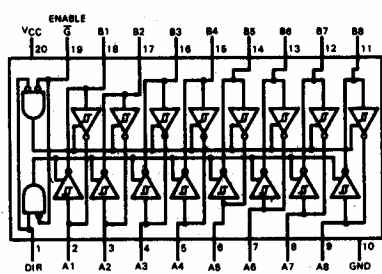
Dual VCO



74LS629

638

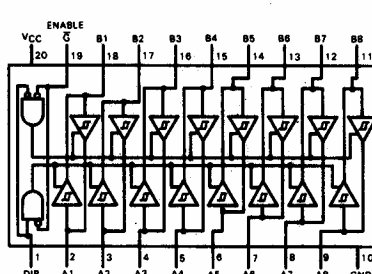
Octal Bus Transceiver with OC (3 state)



74LS638 74F638
74ALS638 74AS638
74ALS1638

639

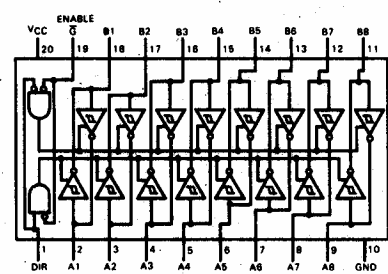
Octal Bus Transceiver with OC (3 state)



74LS639 74F639 74ALS1639
74ALS639 74AS639

640

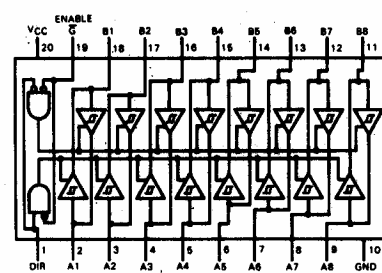
Octal Bus Transceiver (3 state)



74LS640 74ALS640 74HC640
74AS640 74S640 74HCT640
74F640

641

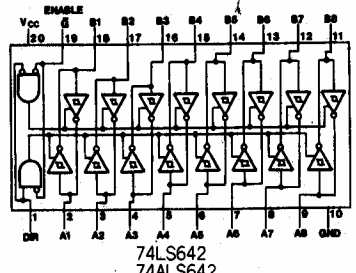
Octal Bus Transceiver with OC



74LS641 74AS641
74ALS641 74ALS1641

642

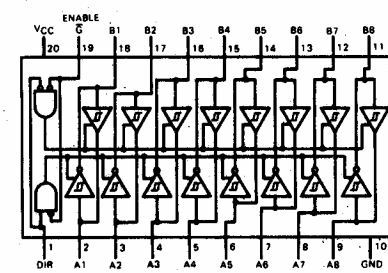
Octal Bus Transceiver with OC



74LS642
74ALS642
74ALS1642
74AS642

643

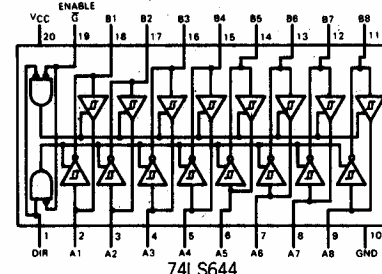
Octal Bus Transceiver (3 state)



74LS643 74S643 74HC643
74ALS643 74F643 74HCT643
74ALS1643

644

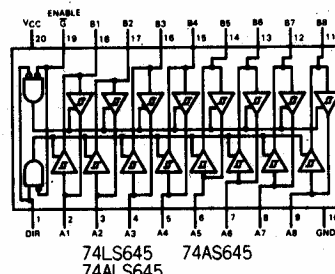
Octal Bus Transceiver with OC



74LS644
74ALS644
74ALS1644
74AS644

645

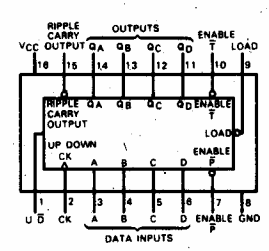
Octal 3 state Bus Transceiver



74LS645 74AS645
74ALS645
74F645
74S645

668

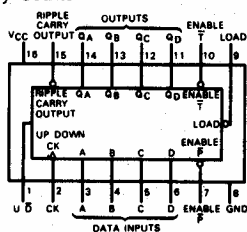
Synchronous Presettable up down Decade Counter



74LS668

669

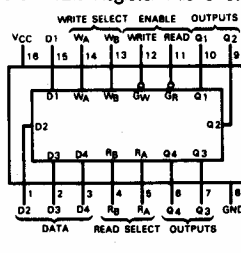
Synchronous Presettable up down Binary Counter



74LS669

670

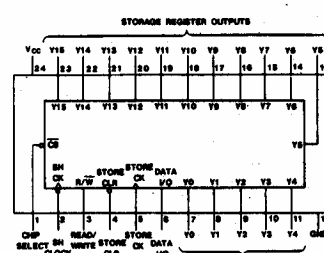
4Word x 4Bit Register File 3 state



74LS670 74HC670
74HCT670

673

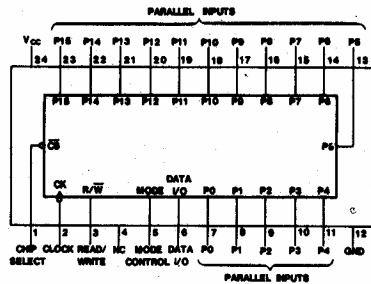
16 Bit Shift Register



74LS673
74F673

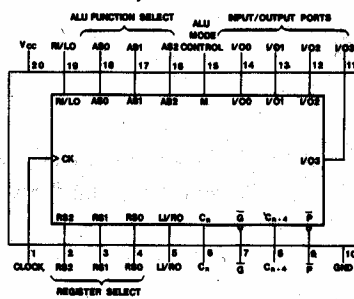
674

16 Bit Shift Register

74LS674
74F674

681

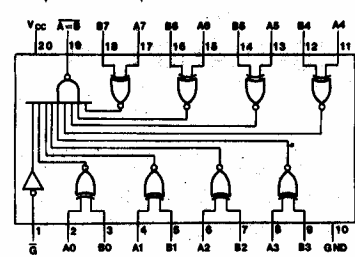
4 Bit Parallel Binary Accumulator



74LS681

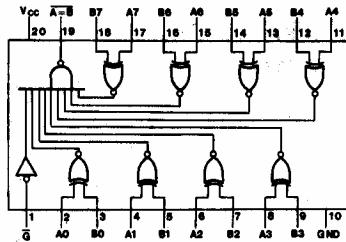
688

8 Bit Equal to Comparator with OC

74LS688 74HC688
74ALS688 74HCT688

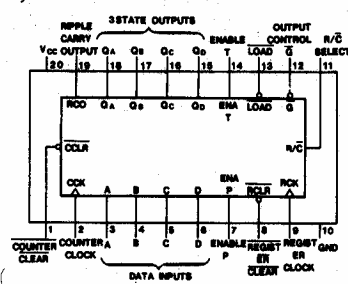
689

8 Bit Equal to Comparator with OC

74LS689
74ALS689

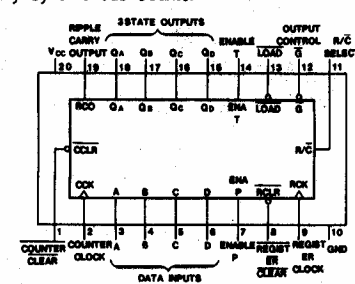
690

BCD Synchronous Counter

74LS690 74HC690
74ALS690

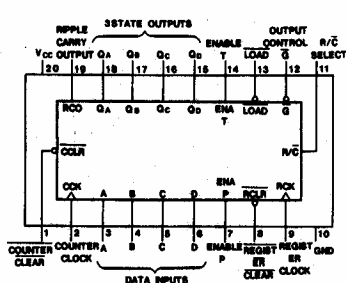
691

Binary Synchronous Counter

74LS691 74HC691
74ALS691

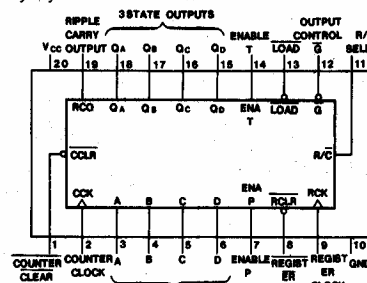
692

BCD Synchronous Counter

74LS692 74HC692
74ALS692

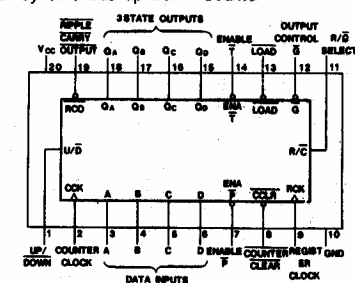
693

Binary Synchronous Counter

74LS693 74HC693
74ALS693

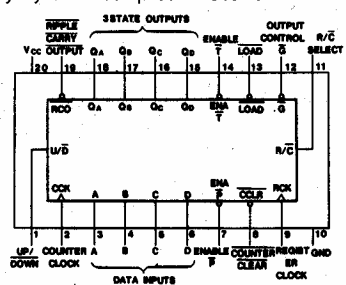
696

BCD Synchronous up down Counter

74LS696 74HC696
74ALS696

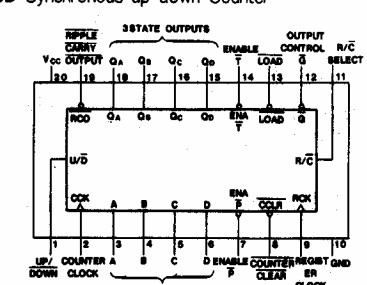
697

Binary Synchronous up down Counter

74LS697 74HC697
74ALS697

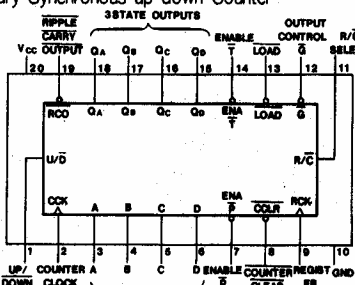
698

BCD Synchronous up down Counter

74LS698 74HC698
74ALS698

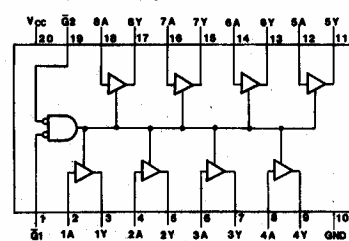
699

Binary Synchronous up down Counter

74LS699 74HC699
74ALS699

795

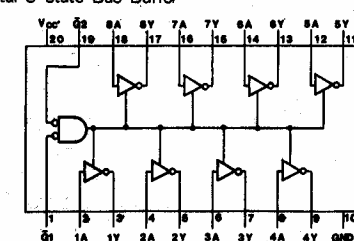
Octal 3 state Bus Buffer



74LS795 81LS95

796

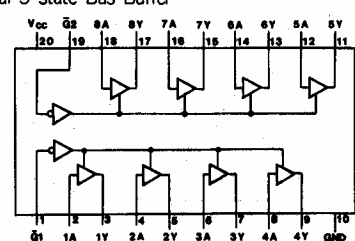
Octal 3 state Bus Buffer



74LS796 81LS96

797

Octal 3 state Bus Buffer



74LS797 81LS97

Absolute Maximum Ratings

SERIES	NOMINAL VALUE OF INPUT PULL-UP RESISTOR (k Ω)	MAXIMUM HIGH-LEVEL CURRENT		MAXIMUM LOW-LEVEL CURRENT		GATES		FLIP-FLOP
		INPUT I_{IH} (μ A)	OUTPUT I_{OH} (μ A)	INPUT I_{IL} (mA)	OUTPUT I_{OL} (mA)	PROPAGATION DELAY TIME (ns)	POWER DISSIPATION (mW)	
94/74	4	40	- 400	- 1.6	16	10	10	DC - 35
54H/74H	2.8	50	- 500	- 2	20	6	22	DC - 50
54LS/74LS	18	20	- 400	- 0.4	4/8	9.5	2	DC - 45
54S/74S	2.8	50	- 1000	- 2	20			
54ALS/74ALS	40	20	- 400	- 0.2	4/8	4	1	DC - 100
54AS/74AS	2.7	200	- 2000	- 2	20	1.5	22	DC - 200
74F		20	- 1000	- 0.6	20	3.0	4	DC - 100

TTL 74 / 74LS / 74ALS / 74F / 74S Series

■ 74LS Series

Name	Function
74LS00	Quad. 2-input Positive NAND Gate
74LS01	Quad. 2-input Positive NAND Gate (Open Collector)
74LS02	Quad. 2-input Positive NOR Gate
74LS03	Quad. 2-input Positive NAND Gate (Open Collector)
74LS04	Hex Inverter
74LS05	Hex Inverter (Open Collector)
74LS08	Quad. 2-input Positive AND Gate
74LS09	Quad. 2-input Positive AND Gate (Open Collector)
74LS10	Triple 3-input Positive NAND Gate
74LS11	Triple 3-input Positive AND Gate
74LS12	Triple 3-input Positive NAND Gate (Open Collector)
74LS13	Dual 4-input Schmitt Trigger Positive NAND Gate
74LS14	Hex Schmitt Trigger Inverter
74LS15	Triple 3-input Positive AND Gate (Open Collector)
74LS20	Dual 4-input Positive NAND Gate
74LS21	Dual 4-input Positive AND Gate
74LS22	Dual 4-input Positive NAND Gate (Open Collector)
74LS26	Quad. 2-input High-Voltage Interface NAND Gate
74LS27	Triple 3-input Positive NOR Gate
74LS28	Quad. 2-input Positive NOR Gate
74LS30	8-input Positive NAND Gate
74LS32	Quad. 2-input Positive OR Gate
74LS33	Quad. 2-input Positive NOR Gate (Open Collector)
74LS37	Quad. 2-input Positive NAND Buffer
74LS38	Quad. 2-input Positive NAND Buffer (Open Collector)
74LS40	Dual 4-input Positive NAND Buffer
74LS42	BCD-to-Decimal Decoder
74LS43	Excess 3-to-Decimal Decoder
74LS47	BCD-to-Seven Segment Decoder/Driver
74LS48	BCD-to-Seven Segment Decoder/Driver
74LS49	BCD-to-Seven Segment Decoder/Driver (Open Collector)
74LS51	Dual 2-wide, 2-input AND-OR-INVERT Gate
74LS54	4-wide, 2-input AND-OR-INVERT Gate
74LS55	2-wide, 4-input AND-OR-INVERT Gate
74LS73	Dual J-K Flip Flop (with Clear)
74LS74	Dual D-type Edge-triggered Flip Flop
74LS75	Quad. Bistable Latch
74LS76	Quad. J-K Flip Flop
74LS77	4-Bit Bistable Latch

Name	Function
74LS78	Dual J-K Flip Flop
74LS83	4-Bit Binary Full Adder (with Fast Carry)
74LS85	4-Bit Magnitude Comparator
74LS86	Quad. 2-input Exclusive-OR Gate
74LS90	Decade Counter
74LS91	8-Bit Shift Register SISO
74LS92	Divide-by-Twelve Counter
74LS93	4-Bit Binary Counter
74LS94	4-Bit Shift Register
74LS95	4-Bit Right Shift Register PIPO
74LS96	5-Bit Shift Register, Dual PIPO
74LS107	Dual J-K Flip Flop
74LS109	Dual J-K Positive Edge-triggered Flip Flop
74LS112	Dual J-K Negative Edge-triggered Flip Flop
74LS113	Dual J-K Negative Edge-triggered Flip Flop
74LS114	Dual J-K Flip Flop
74LS122	Retriggerable Monostable Multivibrator
74LS123	Dual Retriggerable Monostable Multivibrator
74LS125	Quad. Bus Buffer Gate, Inverted 3-state
74LS126	Quad. Bus Buffer Gate, Noninverted 3-state
74LS128	Quad. 2-input NOR 75-ohm Line Driver
74LS132	Quad. 2-input Positive NAND Schmitt Trigger
74LS133	13-input Positive NAND Gate
74LS136	Quad. 2-input Exclusive-OR Gate (Open Collector)
74LS137	3-to-8 Line Decoder/Demultiplexer
74LS138	3-to-8 Line Decoder
74LS139	Dual 2-to-4 Line Decoder/Demultiplexer
74LS145	BCD-to-Decimal Decoder/Driver
74LS148	8-to-3 Line Priority Encoder
74LS151	8-Bit Data Selector/Multiplexer (with Strobe)
74LS152	1-of-8 Data Selector/Multiplexer
74LS153	Dual 4-to-1 Line Data Selector/Multiplexer
74LS154	4-to-16 Line Decoder/Demultiplexer
74LS155	Dual 2-to-4 Line Decoder
74LS156	Dual 2-to-4 Line Decoder (Open Collector)
74LS157	Quad. 2-to-1 Line Data Selector/Multiplexer
74LS158	Quad. 2-to-1 Line Data Selector/Multiplexer
74LS160	Synchronous Decade Counter (with Direct Clear)
74LS161	Synchronous Binary Counter (with Direct Clear)
74LS162	Synchronous Decade Counter (Synchronous Clear)

Name	Function
74LS163	Synchronous Binary Counter (Synchronous Clear)
74LS164	8-Bit SIPO Shift Register (Asynchronous Clear)
74LS165	8-Bit PISO Shift Register
74LS166	Parallel Load 8-Bit Shift Register
74LS169	Synchronous 4-Bit Up/Down Binary Counter
74LS170	4x4 Register File (Open Collector)
74LS173	4-Bit D-type Register, 3-State
74LS174	Hex D-type Flip Flop (with Clear)
74LS175	Quad, D-type Flip Flop (with Clear)
74LS181	4-Bit Arithmetic Logic Unit/Function Generator
74LS190	Synchronous Up/Down Counter, Decade
74LS191	Synchronous Up/Down Counter, Binary
74LS192	Synchronous Up/Down Counter, Decade
74LS193	Synchronous Up/Down Counter, Binary
74LS194	4-Bit PIPO Bidirectional Shift Register
74LS195	4-Bit PIPO Shift Register
74LS196	4-Stage Presetable Ripple Counter, Decade
74LS197	4-State Presetable Ripple Counter, Binary
74LS221	Dual Monostable Multivibrator
74LS224	16Wx4B Asynchronous FIFO Memory
74LS240	Octal Buffer/Driver/Receiver (Inverted 3-State)
74LS241	Octal Buffer/Driver/Receiver (Noninverted 3-State)
74LS242	Quad. Bus Transceiver, 3-State
74LS243	Quad. Bus Transceiver, 3-State
74LS244	Octal Buffer/Driver/Receiver (Noninverted 3-State)
74LS245	Octal Bus Transceiver (Noninverted 3-State)
74LS247	BCD-to-Seven Segment Decoder/Driver
74LS248	BCD-to-Seven Segment Decoder/Driver
74LS249	BCD-to-Seven Segment Decoder/Driver
74LS251	8-Bit Data Selector/Multiplexer, 3-State
74LS253	Dual 4-to-1 Line Data Selector/Multiplexer, 3-State
74LS257	Quad. 2-to-1 Line Data Selector/Multiplexer, 3-State
74LS258	Quad. 2-to-1 Line Data Selector/Multiplexer, 3-State
74LS259	8-Bit Addressable Latch
74LS260	Dual 5-input Positive NOR Gate
74LS266	Quad. 2-input Exclusive-NOR Gate (Open Collector)
74LS273	8-Bit D-type Shift Register
74LS279	Quad. S-R Latch
74LS280	9-Bit Odd/Even Parity Generator/Checker
74LS283	4-Bit Binary Full Adder (with Fast Carry)
74LS290	Decade Counter
74LS293	4-Bit Binary Counter
74LS295	4-Bit Shift Register, 3-State
74LS297	Digital Phase-Locked Loop Filter
74LS298	Quad. 2-input Multiplexer
74LS299	8-Bit Universal Shift/Storage Register
74LS322	8-Bit Shift Register with Single Ended
74LS323	8-Bit Universal Shift/Storage Register
74LS352	Dual 4-to-1 Data Selector/Multiplexer
74LS353	Dual 4-to-1 Data Selector/Multiplexer
74LS365	Hex Bus Buffer/Driver, 3-State
74LS366	Hex Bus Buffer/Driver, 3-State
74LS367	Hex Bus Buffer/Driver, 3-State
74LS368	Hex Bus Buffer/Driver, 3-State
74LS373	Octal Transparent Latch, 3-State
74LS374	Octal D-type Flip Flop, 3-State
74LS375	4-Bit Bistable Latch
74LS377	Octal D-type Flip Flop
74LS378	Hex Parallel D Register
74LS386	Quad. 2-input Exclusive-OR Gate

Name	Function
74LS390	Dual Decade Counter
74LS393	Dual 4-Bit Binary Counter
74LS395	4-Bit Shift Register, 3-state
74LS490	Dual 4-Bit Decade Counter
74LS540	Octal Buffer/Driver, 3-State
74LS541	Octal Buffer/Driver, 3-State
74LS573	Octal D-type Transparent Latch
74LS574	Octal D-type Edge-triggered Flip Flop
74LS590	8-Bit Binary Counter/Register
74LS597	8-Bit Latch/Shift Register
74LS612	Memory Mappers
74LS620	Octal Bus Transceiver
74LS623	Octal Bus Transceiver
74LS624	Voltage-Controlled Oscillator
74LS629	Voltage-Controlled Oscillator
74LS640	Octal Bus Transceiver
74LS641	Octal Bus Transceiver (Open Collector)
74LS642	Octal Bus Transceiver
74LS645	Octal Bus Transceiver, 3-State
74LS646	Octal Bus Transceiver/Register, 3-State
74LS668	Synch. Presetable Decade Up/Down Counter
74LS669	Synch. Presetable Binary Up/Down Counter
74LS670	4x4 Register Files, 3-State
74LS688	8-Bit Magnitude Comparator

■ 74ALS Series

Name	Function
74ALS00	Quad. 2-input Positive NAND Gate
74ALS02	Quad. 2-input Positive NOR Gate
74ALS04	Hex Inverter
74ALS08	Quad. 2-input Positive AND Gate
74ALS10	Triple 3-input Positive NAND Gate
74ALS20	Dual 4-input Positive NAND Gate
74ALS27	Triple 3-input Positive NOR Gate
74ALS30	8-input Positive NAND Gate
74ALS37	Quad. 2-input Positive NAND Buffer
74ALS38	Quad. 2-input Positive NAND Buffer (Open Collector)
74ALS40	Dual 4-input Positive NAND Buffer
74ALS74	Dual D-type Edge-triggered Flip Flop
74ALS86	Quad. 2-input Exclusive-OR Gate
74ALS112	Dual J-K Negative Edge-triggered Flip Flop
74ALS133	13-input Positive NAND Gate
74ALS138	3-to-8 Line Decoder
74ALS151	8-Bit Data Selector/Multiplexer
74ALS153	Dual 4-to-1 Line Data Selector/Multiplexer
74ALS161	Synchronous Binary Counter
74ALS163	Synchronous Binary Counter
74ALS174	Hex D-type Flip Flop
74ALS175	Quad. D-type Flip Flop
74ALS191	Synchronous Up/Down Counter, Binary
74ALS192	Synchronous Up/Down Counter, Decade
74ALS193	Synchronous Up/Down Counter, Binary
74ALS240	Octal Buffer/Driver/Receiver (Inverted 3-State)
74ALS241	Octal Buffer/Driver/Receiver (Noninverted 3-State)
74ALS244	Octal Buffer/Driver/Receiver (Noninverted 3-State)
74ALS245	Octal Bus Transceiver (Noninverted 3-State)
74ALS251	8-Bit Data Selector/Multiplexer, 3-State

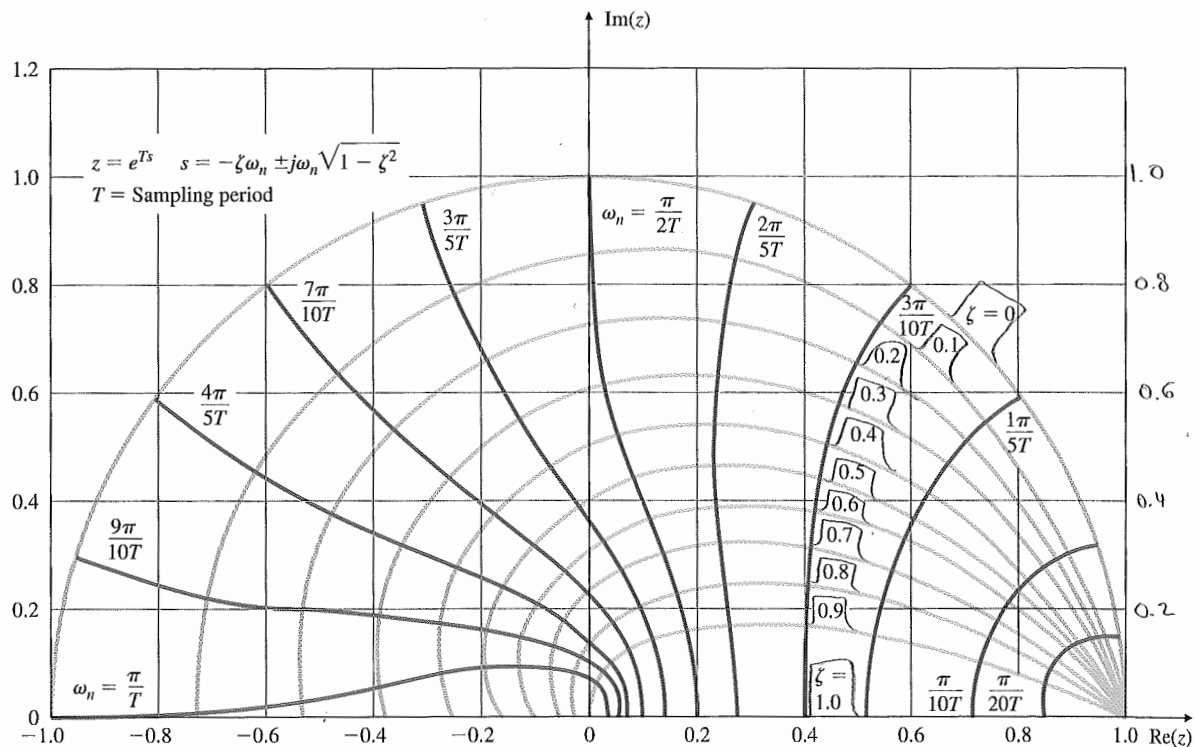


Figure 8.4